Segmented Recurrent Transformer With Cubed 3-D-Multiscanning Strategy for Hyperspectral Image Classification

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***Abstract*— This study introduces an innovative approach to hyperspectral imaging (HSI) classification by integrating convolution, recurrence, and self-attention mechanisms in a 3-D configuration. We address several challenges such as the 1) disruption of spectral continuity by traditional dimensionality reduction methods like PCA, 2) the overlooking of band-to-band continuous features in existing spatial-only 2-D multiscanning strategy, and 3) the limitations in model design by simply cascading recurrent neural networks (RNNs) with transformers for HSI analysis. Our solution involves three core components: 1) subband grouping with group-wise convolution for refined dimension reduction; 2) a novel cubed 3-D-multiscanning technique enabling thorough multidirectional analysis in both spectral and spatial domains; and 3) the development of a Cubic-Net framework with a specially designed segmented recurrent transformer (SRT). This SRT is tailored to effectively utilize spectral continuity along with spatial contextual features, overcoming common sequential data analysis challenges seen in RNNs and transformers. Furthermore, our feature fusion strategy successively integrates "short-term" and "long-term" SRT features, thereby enhancing the model's ability to process both spectral and spatial features effectively. Experimental results from three public HSI datasets indicate our method's improved performance over existing baselines and state-of-the-art methods. This research offers a new perspective on 3-D sequential HSI classification.**

***Index Terms*— Convolutional neural networks (CNNs), deep learning, hyperspectral image classification, multiscanning strategy, recurrent neural networks (RNNs), transformer.**

I. Introduction

HYPERSPECTRAL imaging (HSI) is an advanced remote sensing (RS) technique that captures images across numerous dense and continuous spectral bands, covering the electromagnetic spectrum from visible to infrared wavelengths. This imaging is often conducted by satellites or unmanned aerial vehicles (UAVs) [1]. Unlike traditional imaging methods, HSI provides detailed spectral information for each pixel, offering a more precise characterization of materials and objects on the Earth's surface [2]. The versatility of this technology has led to its application in diverse fields such as agriculture [3], environmental monitoring [4], geology [5], urban planning [6], and defense and security [7]. Given these wide-ranging applications, the task of HSI classification, which involves assigning a label to each pixel, is of paramount importance. High accuracy in HSI classification not only enhances the quality of these applications but is also a focal point of extensive research in the RS domain [8].

In recent years, the development of deep learning-based models for HSI classification tasks has been extensive [9], [10]. These models generally adopt a patch-wise learning framework, which processes a local patch around each pixel to assign it a semantic label [11]. Convolutional neural networks (CNNs) [12], [13], [14], [15], [16], [17], recurrent neural networks (RNNs) [18], [19], [20], [21], [22], [23], and vision transformers (ViTs) [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], among others, have been widely implemented in this context. While these methods have achieved significant results, there remain several issues that require careful consideration.

First, the high dimensionality of HSI often necessitates dimensionality reduction techniques to conserve computational resources and mitigate the Hughes phenomenon [36]. Principal component analysis (PCA) is a common method in this regard, used extensively in various studies [37], [38], [39], [40], [41], [42]. However, given that spectral information in HSI is densely recorded, akin to video frames, the indiscriminate use of PCA may disrupt the continuity of spectral data. This disruption poses a challenge, especially when integrating with RNNs or transformers that depend on sequential data analysis. To maintain spectral continuity, alternative methods like band degrouping [21], [24], depth-wise 2D-CNNs [43], [44], [45], and lightweight 3D-CNNs [46], [47], [48] have been explored. Band degrouping processes spectral bands locally for individual feature representation, while depth-wise 2D-CNNs distribute spectral features across channels, reducing the feature map's channel count. Three-dimensional-CNNs, on the other hand, use consistent spectral kernels and strides for a compressed local channel feature map. While these approaches have shown improvements, they may inadvertently amalgamate features due to the uniform

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division of the spectral domain, as spectral properties can vary significantly across bands. This potential issue can be assessed by computing the Pearson correlation coefficient between bands. Typically, these methods create equal-sized subgroups for local feature aggregation, which could lead to the grouping of uncorrelated adjacent bands, thereby fusing them. This may not effectively handle spectral dependencies on both local and global scales. Therefore, properly managing spectral continuity features remains a crucial challenge.

Second, previous works on the 2-D-multiscanning strategy [49], [50], [51] process an HSI patch using various scan orders to combine complementary spatial contextual features, with each scan order acting as a positional forward pass for sequential models. However, the 2-D-multiscanning strategy only considers positioning in the spatial domain, completely ignoring the spectral continuity feature. As previously mentioned, the spectral information in an HSI is recorded densely and continuously, exhibiting a high correlation among band-to-band ordering. Consequently, the existing 2-D version of the multiscanning strategy is insufficient for comprehending the HSI's spectral dependencies, potentially leading to degradation in model training. This limitation is analogous to the difference between 2D-CNN and 3D-CNN, where the latter typically yields better results due to its ability to handle spectral information through both spectral and spatial convolution. Although Song et al. [52] proposed a multidirectional scan for spectral and spatial features, this approach separates the spectral and spatial processes, resulting in a disconnection between spectral and spatial features. In summary, it is necessary to extend the 2-D version of the multiscanning strategy to a 3-D version that can effectively scan and correlate spectral and spatial features for subsequent sequential models.

Third, interestingly, some studies [53], [54], [55], [56], [57] have highlighted the effectiveness of combining RNNs with Transformers in certain specific tasks. This combination is particularly valuable for capturing structural representations [58], [59], [60], [61], [62] and positional embeddings in input sequences [63], which are areas where transformers' self-attention mechanisms are less adept. Moreover, researches [64], [65] indicate that the learning representations of Transformers and RNNs are complementary. This insight led to its application in HSI classification. Zhou et al. [51] proposed a cascaded recurrent-transformer (RT) block that integrates RNNs with a pixel transformer (PT) to address both ordering bias and pixel-level self-attention. However, despite its inspiration, this method shares certain limitations with other approaches [66], [67], [68], [69], [70] that also use a combination of RNNs and transformers. Primarily, its cascading or parallel structure results in increased computational complexity and burden by simply stacking several layers and may not fully harness the potential synergies of RNNs and transformers, potentially leading to suboptimal capture of the spatial–spectral dynamics in HSI data. Furthermore, the aforementioned models are generally constrained by 2-D input, processing each row or column (i.e., a vector) sequentially. This approach significantly limits their flexibility when dealing with higher-dimensional input shapes (i.e., each step is an image or a tensor). There is, therefore, a need for more nuanced and efficient methods that better integrate the recurrence capabilities of RNNs with the self-attention mechanisms of transformers, ensuring a more effective and computationally efficient approach to HSI data analysis.

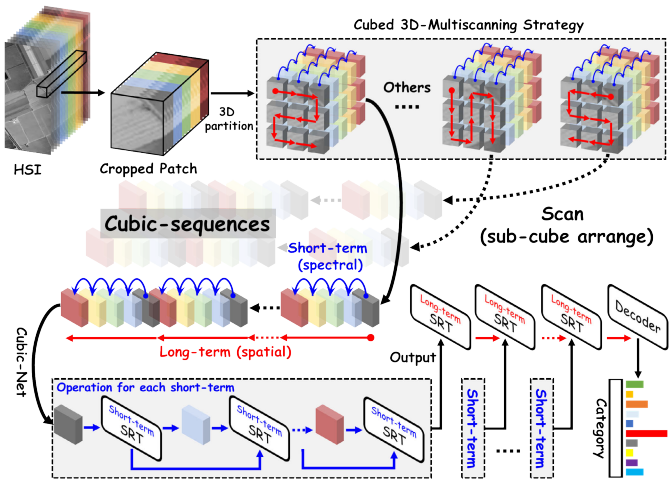


Fig. 1. Concept of proposed method. It begins with cropping patches from the original HSI, which are then partitioned into a 3-D hyper-cube. Subcubes undergo a 3-D-multiscanning process, creating multiple cubic-sequences that intercross different spectral–spatial features. SRT is designed with two types: "short-term" for handling spectral continuous features, and "long-term" for managing spatial contextual features. A decoder is prepared for categorization.

In light of the three scenarios discussed previously, we propose a novel approach to processing patch-cubes by integrating convolution, recurrence, and self-attention mechanisms. It is comprised of three key components: 1) subband grouping with group-wise convolution, designed to aggregate spectral features unequally for more nuanced dimension reduction; 2) a cubed 3-D-multiscanning technique for multidirectional cubic positioning within the spectral and spatial domains; and 3) a Cubic-Net framework that incorporates a novel designed segmented RT (SRT). The SRT is inherently engineered to leverage the strengths of both recurrence and self-attention. We also introduce an advanced feature fusion strategy that combines "short-term" and "long-term" SRT features across the spectral and spatial dimensions in a sequential manner, enhancing the model's ability to generate features. The proposed multiscanning feature fusion method further integrates these scanning features in a weighted fashion, significantly improving the model's generative capacity. The brief concept of this study is depicted in Fig. 1.

The contributions of this study are summarized as follows.

1) This study is the first to unify convolution, recurrence, and self-attention, particularly through the Cubic-Net with its specially designed SRT for unified spectral–spatial cubic-sequence processing, demonstrating its effectiveness in HSI classification tasks.
2) We introduce a novel subband grouping and group-wise convolution approach for unequal spectral feature aggregation, proving its efficiency in feature aggregation and dimension reduction.
3) The extended 3-D multiscanning strategy demonstrates improved handling of spectral feature continuity and

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enhances the model's generative capabilities, especially with limited data samples.

The rest of the article is organized as follows. Section II details the methodology, Section III discusses the experimental setup and results, and Section IV concludes the study.

II. Methodology

This section first presents the methodological overview of the proposed method (Section II-A) briefly. After that, it presents the detailed contents: subband grouping strategy (Section II-B), group-wise convolution (Section II-C), cubed 3-D-multiscanning for cubic positioning (Section II-D), Cubic-Net framework (Section II-E), and multiscanning feature fusion (Section II-F). The overall structure and the pseudo algorithm are shown in Fig. 2 and Algorithm 1, respectively.

A. Methodological Overview of Proposed Method

Our method comprises several key components as follows.

1) A subband grouping strategy for an uneven spectral division, which groups highly correlated band images locally and adjacently.
2) Group-wise convolutions for unifying subband groups and reducing dimensions.
3) A 3-D cubic partition along the spatial and spectral domains, forming a hyper-cube structure from the unified and reconcatenated image.
4) A 3-D-multiscanning strategy to generate multiple directional cubic-sequences.
5) The Cubic-Net, equipped with proposed SRTs, processes these cubic-sequences both spatially (long-term) and spectrally (short-term).
6) A bi-directional feature aggregation and a weighted multiscanning feature fusion strategy are employed for subsequent decoding.

Algorithm 1 Pseudo Procedure of the Proposed Method

Input: Input HSI data $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, ground-truth $\mathbf{L} \in \mathbb{R}^{H\times W}$; patch size p; fixed training sample size.
Output: Predicted labels of the testing set.
Initialization: Optimizer: AdamW; Learning rate: 0.001; Criterion: CrossEntropy; Epochs: 200; Batch size: 100.
Create sample patches from $\mathbf{X}$, divide into training and test datasets.
Calculate inter-band cross-correlation matrix $\mathbf{r}_{all}$ for $\mathbf{X}$ (Eq. 1).
Slice patch $\mathbf{X}^{(i,j)}$ into sub-groups based on $\mathbf{r}_{all}$ (Eq. 5).
Apply group-wise CNNs on each sub-band group $\mathbf{X}_s^{(i,j)}$ for unifying and reduing spectral dimension (Eq. 6).
Concatenate unified sub-bands into $\tilde{\mathbf{X}}^{(i,j)}$ (Eq. 7).
Perform 3D cubic partitioning for $\tilde{\mathbf{X}}^{(i,j)}$, denoted as hyper-cube structure.
Scan sub-cubes using 3D-multiscanning (Eqs. 12, 13).
Generate cubic-sequences $\{\mathbf{S}_m^{(i,j)}, m = 1, 2, \ldots, M\}$, where M denotes the total number of used scan orders.
Cubic-Net Framework:
for $m = 1$ **to** M in $\mathbf{S}_m^{(i,j)}$ **do**
    Segment cubic-sequence $\mathbf{S}_m^{(i,j)}$ into sub-sequences $\{\mathbf{S}_{m,N_s}^{(i,j)}\}$, where N_s equals total number of sub-sequences (Eq. 14a).
    Process each sub-sequence with short-term SRT for output feature $\mathbf{O}_{m,t}^{(i,j)}$ (Eq. 28).
    Concatenate short-term outputs as $\tilde{\mathbf{O}}_m^{(i,j)}$, feed into long-term SRT.
    return $\mathbf{F}_m^{(i,j)}$ (Eq. 35).
end for
Perform bi-directional feature aggregation to obtain $\tilde{\mathbf{F}}_e^{(i,j)}$ (Eq. 37).
Execute weighted multiscanning feature fusion (Eq. 38).
Use decoder with loss function for training (Eq. 39).

B. Subband Grouping Strategy

Unlike traditional RGB images with only three bands, the HSI can encompass hundreds of contiguous bands across a broad wavelength range. This high spectral resolution captures subtle spectral differences, leading to high correlation and minimal variation between neighboring bands. Such similarity results in data redundancy and heavy computational load.

To tackle these challenges, many studies utilize PCA for dimensionality reduction. However, PCA's tendency to disrupt spectral continuity can pose problems, particularly when applying sequential operations like depth-wise 2D-CNNs, sliding kernels of 3D-CNNs, or the step-wise recurrence of RNNs. To counteract this, some researchers have suggested segmenting spectral bands into groups [71], [72]. It aggregates adjacent bands to reduce local redundancy, while globally integrating features from all groups to ensure complementary information. While some studies have divided spectral bands uniformly into equal-sized segments, they overlook the variable correlations among adjacent bands in the spectral domain of HSI.

The cross-correlations between bands in an HSI ($\mathbf{X} \in \mathbb{R}^{H\times W\times C}$), can be quantitatively assessed by calculating the normalized Pearson's correlation coefficient as

$$r_{xy} = \frac{\sum_{i=1}^{H\times W}(B_{xi} - \bar{B}_x)(B_{yi} - \bar{B}_y)}{\sqrt{\sum_{i=1}^{H\times W}(B_{xi} - \bar{B}_x)^2 \sum_{i=1}^{H\times W}(B_{yi} - \bar{B}_y)^2}}. \quad (1)$$

Here, r_{xy} denotes the correlation coefficient between the band images $\mathbf{B}_x$ and $\mathbf{B}_y$, where $\mathbf{B}_x, \mathbf{B}_y \in \mathbb{R}^{HW\times 1}$. The pixel values at the ith position in $\mathbf{B}_x$ and $\mathbf{B}_y$ are B_{xi} and B_{yi}, with $\bar{B}_x$ and $\bar{B}_y$ as their mean values. By calculating the cross correlation for all band pairs, we can obtain the whole correlation matrix $\mathbf{r}_{\text{all}} \in \mathbb{R}^{C\times C}$, which is vital for understanding interband relationships. This informs a more sophisticated view of band division, as illustrated in Fig. 3.

Hence, rather than employing a uniform division, we introduce a subband grouping strategy that relies on the interband cross correlation matrix $\mathbf{r}_{\text{all}}$, partitioning the bands into several uneven subband groups manually as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_S] \quad (2)$$

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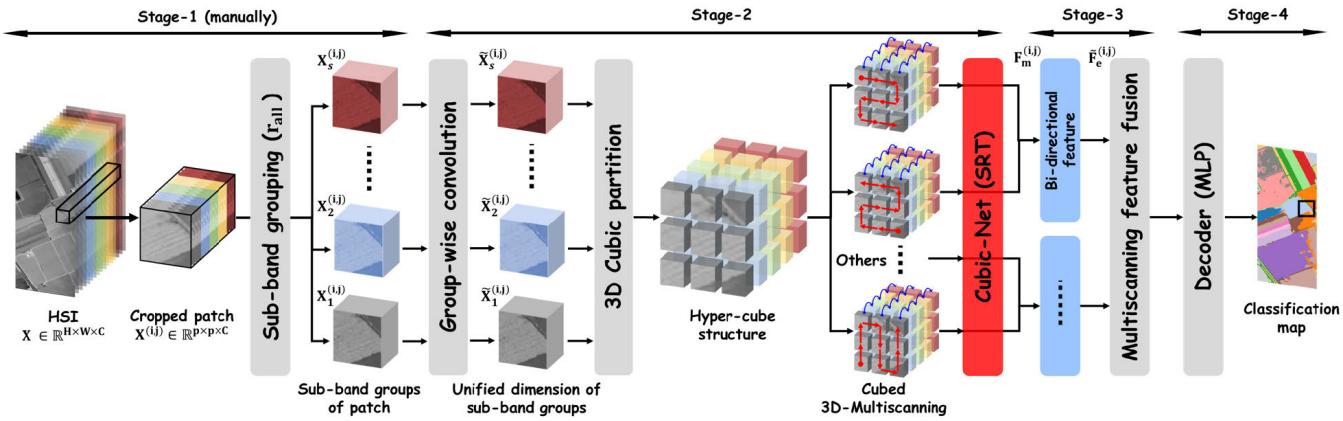


Fig. 2. Overall structure. Stage-1: structure preparation (manually). This stage involves preparing the necessary structures for the latter procedure, such as the subband grouping strategy. Stage-2: cubed 3-D-multiscanning based cubic-sequence processing. In this stage, group-wise convolution and 3-D cubic partition are first conducted and then 3-D-multiscanning oriented Cubic-Nets are applied to process the multiple cubic-sequences. Stage-3: bi-directional feature generation with weighted fusion. This stage focuses on unifying reversed scanning features in a bi-directional manner, after that the weighted fusion from multiscanning features is implemented. Stage-4: decoder with classification. The final stage involves decoding and classification.

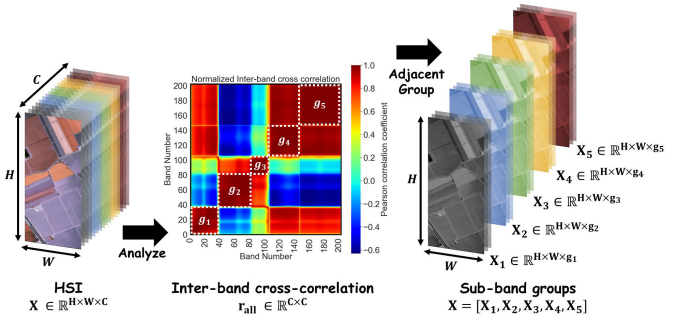


Fig. 3. Subband grouping strategy groups adjacent spectral bands into nonuniform segments according to interband cross correlation analysis, with the diagonal direction in the visualization indicating these correlations. In this case, the total number of bands $C = g_1 + g_2 + g_3 + g_4 + g_5$.

where $\mathbf{X}_s \in \mathbb{R}^{H\times W\times g_s}$. g_s represents the number of bands in the sth subgroup. S is the total number of subband groups. Therefore, we accomplish the spectral partitioning, and these subband groups are then processed independently later.

C. Group-Wise Convolution

Designed as a patch-wise framework, the input for the neural model is a cropped patch from the original HSI. For one cropped HSI patch $\mathbf{X}^{(i,j)}$ centered at $\mathbf{x}^{(i,j)}$ with patch size p, an odd number, the patch can be represented as

$$\mathbf{X}^{(i,j)} = \left\{ \mathbf{x}^{(i+\alpha, j+\beta)} \,\middle|\, \alpha, \beta = -\frac{p-1}{2}, \ldots, \frac{p-1}{2} \right\} \tag{3}$$

where $\alpha, \beta \in \mathbb{Z}$ and $i+\alpha$ and $j+\beta$ record the position.

Taking $p = 9$ as an example, the cropped HSI patch $\mathbf{X}^{(i,j)}$ can be represented as

$$\mathbf{X}^{(i,j)} = \begin{pmatrix} \mathbf{x}^{(i-4,j-4)} & \mathbf{x}^{(i-4,j-3)} & \cdots & \mathbf{x}^{(i-4,j+4)} \\ \mathbf{x}^{(i-3,j-4)} & \mathbf{x}^{(i-3,j-3)} & \cdots & \mathbf{x}^{(i-3,j+4)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}^{(i+4,j-4)} & \mathbf{x}^{(i+4,j-3)} & \cdots & \mathbf{x}^{(i+4,j+4)} \end{pmatrix} \tag{4}$$

where $4 \le i \le H-5$, $4 \le j \le W-5$, and $\mathbf{X}^{(i,j)} \in \mathbb{R}^{p\times p\times C}$. The center pixel $\mathbf{X}^{(i,j)}[4,4]$ is $\mathbf{x}^{(i,j)}$, and each pixel is represented by a vector in $\mathbb{R}^C$.

Hence, the cropped HSI patch can be also partitioned into several subband groups, referring to (2), as

$$\mathbf{X}^{(i,j)} = \left[\mathbf{X}_1^{(i,j)}, \mathbf{X}_2^{(i,j)}, \ldots, \mathbf{X}_S^{(i,j)} \right] \tag{5}$$

where $\mathbf{X}_s^{(i,j)} \in \mathbb{R}^{p\times p\times g_s}$.

For each subband patch $\mathbf{X}_s^{(i,j)}$, we design a specific CNNs' module (i.e., Conv2D, batch normalization, ReLU, and residual connection) with input channel g_s to perform the convolution and output feature maps with dimension d

$$\tilde{\mathbf{X}}_s^{(i,j)} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathbf{X}_s^{(i,j)} \star \mathbf{K}_s^{(i,j)}\right)\right) + \mathbf{X}_s^{(i,j)} \tag{6}$$

where $\mathbf{K}_s^{(i,j)} \in \mathbb{R}^{k_1\times k_2\times g_s\times d}$ represents the sth convolutional kernel for the sth subband patch, and $\star$ denotes the convolution operation. Particularly, the CNNs across each subband patch do not share parameters to avoid unexpected feature mixtures from different subband groups, which is named group-wise convolution. The detailed design of group-wise convolution is illustrated in Fig. 4.

Hence, outputs from the group-wise convolutions are independent and concatenated into a new representation as

$$\tilde{\mathbf{X}}^{(i,j)} = \mathrm{Concat}\left(\tilde{\mathbf{X}}_1^{(i,j)}, \tilde{\mathbf{X}}_2^{(i,j)}, \ldots, \tilde{\mathbf{X}}_S^{(i,j)}\right) \tag{7}$$

where $\tilde{\mathbf{X}}_s^{(i,j)} \in \mathbb{R}^{p_1\times p_2\times d}$. And, $\tilde{\mathbf{X}}^{(i,j)} \in \mathbb{R}^{p_1\times p_2\times D}$, where $D = S \times d$. The spatial size of $p_1 \times p_2$ is obtained by

$$p_1 = \left(\frac{p + 2\times p_d - k_1}{s_1} + 1 \right) \tag{8}$$

$$p_2 = \left(\frac{p + 2\times p_d - k_2}{s_2} + 1 \right) \tag{9}$$

where s_1 and s_2 represent the stride on the spatial domain and p_d represents the padding size. k_1 and k_2 represent the convolutional kernel size. Generally, $s_1 = s_2 = 1$, $p_d = 1$, and $k_1 = k_2 = 3$. Hence, the subband patches are transformed into a uniform channel dimension for subsequent processing.

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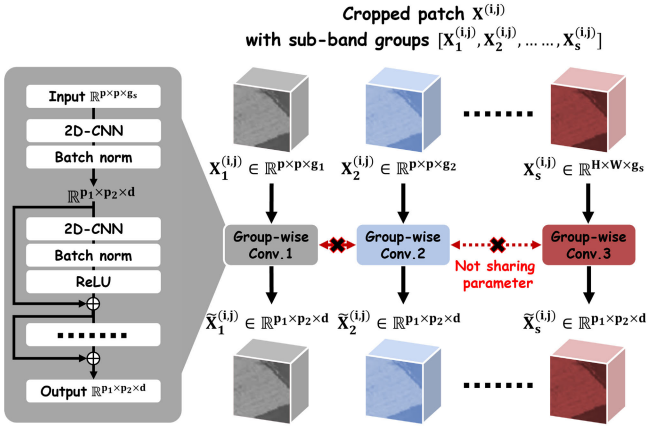


Fig. 4. Subband groups are created from the patch and partitioned according to their spectral characteristics. Each subband group undergoes processing with specialized convolutions for subband unification and reduction. Importantly, convolution parameters are unique to each subband group and are not shared. The output channels of these convolutions are uniformly configured.

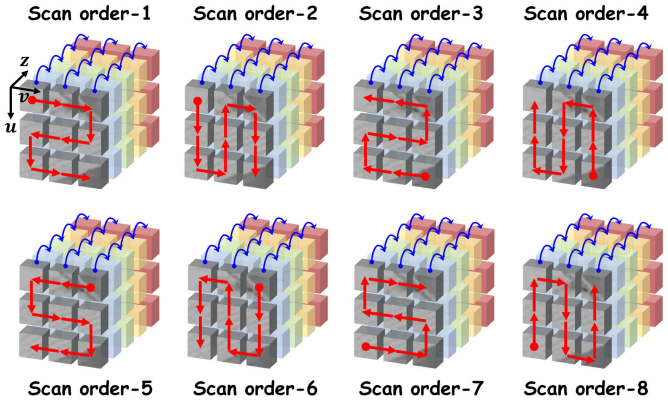


Fig. 5. Three-dimensional-multiscanning strategy. It involves scanning the subcube in multiple directions to capture comprehensive spatial and spectral information. The blue lines represent the scanning in the spectral domain, which involves traversing through the spectral bands. The red lines indicate the scanning in the spatial domain, highlighting the traversal across spatial dimensions.

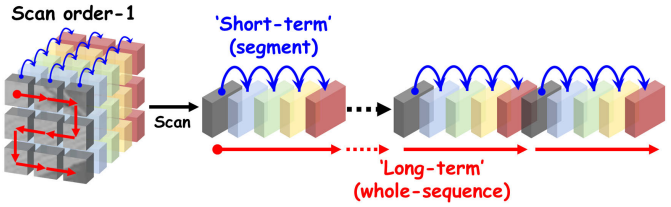


Fig. 6. Example of flattening ordering under scan order-1. In this context, "scan" refers to the process of rearranging the hyper-cube into a cubic-sequence. Specifically, "short-term" denotes segments that contain spectral continuous features. "Long-term," represents the spatial contextual features.

Our approach diverges from the commonly employed depth-wise convolution by adapting the uneven group size to optimize spectral feature generation. Traditional convolutional methods typically rely on fixed stride or group sizes as hyperparameters; however, this rigidity may not be ideal for the nuanced aggregation of spectral features in HSI.

D. Cubed 3-D-Multiscanning for Cubic Positioning

Previous studies [49], [50], [51] have validated the effectiveness of using multiple scanning techniques to reorganize HSI patches into various feature sequences, thereby improving feature learning and classification results. These methods, however, have largely focused on the spatial domain of images, neglecting the inherent band-to-band dependencies present in HSIs—dependencies that are crucial due to the high spectral resolution and continuity akin to frames in a video.

In response, Fig. 5 graphically depicts our proposed 3-D-multiscanning strategy, explicitly accounting for spectral dependencies. This advanced multiscanning technique reconfigures the subcubes into multiple cubic sequences for enhanced feature extraction. The model then processes these sequences for further feature learning.

1) Cubic Partition: After obtaining the image $\tilde{\mathbf{X}}^{(i,j)}$ within a space of $\mathbb{R}^{p_1\times p_2\times D}$, we divide it into several nonoverlapping subcubes, each of the size $\mathbb{R}^{p_c\times p_c\times d_c}$. The total count of these subcubes, N_c, is determined by the ratio of its dimensions to those of a subcube, expressed as

$$N_c = \frac{p_1 \times p_2 \times D}{p_c \times p_c \times d_c}. \tag{10}$$

Hence, the partitioned $\tilde{\mathbf{X}}^{(i,j)}$ is conceptualized as a "hyper-cube" made up of a series of smaller subcubes $\tilde{\mathbf{x}}_{u,v,z}^{(i,j)}$, which are containing both "local" spatial and spectral information

$$\tilde{\mathbf{X}}^{(i,j)} = \left\{ \tilde{\mathbf{x}}_{u,v,z}^{(i,j)} \,\middle|\, u = 0 \sim n_h, v = 0 \sim n_w, z = 0 \sim n_f \right\} \tag{11}$$

where u, v, and z are integer indices within the "hyper-cube," representing the position of each subcube along the height, width, and depth dimensions, respectively, with $n_h = (p_1/p_c) - 1$, $n_w = (p_2/p_c) - 1$, and $n_f = (D/d_c) - 1$.

2) Cubic Positioning With 3-D-Multiscanning: After the 3-D partition into subcubes, we proceed to organize them into various cubic-sequences, denoted as $\{\mathbf{S}_m^{(i,j)}, m = 1, 2, \ldots, M\}$. Each sequence is a flattened representation in $\mathbb{R}^{N_c\times D_c}$ space, where D_c embodies the cumulative dimension of a subcube, calculated as $p_c \times p_c \times d_c$. Here, m signifies the scan order index within the set of scanning sequences. The specific arrangement of these sequences is guided by Fig. 5.

For instance, considering scan order-1 as illustrated in Fig. 6, the organization commences at $\tilde{\mathbf{x}}_{0,0,0}^{(i,j)}$, proceeding along the spectral domain to $\tilde{\mathbf{x}}_{0,0,n_f}^{(i,j)}$, as indicated by the blue trajectory. Upon reaching the end of the spectral scan, the sequence transitions to the next spatial domain position at $v = 1$. This scanning process is repeated for all subcubes, thereby converting the entire "hyper-cube" into a single, linear cubic-sequence in accordance with the scan order-1 as follows:

$$\begin{aligned}
&\tilde{\mathbf{x}}_{0,0,0}^{(i,j)} \rightarrow \tilde{\mathbf{x}}_{0,0,1}^{(i,j)} \rightarrow \cdots \rightarrow \tilde{\mathbf{x}}_{0,0,n_f}^{(i,j)} \\
&\rightarrow \tilde{\mathbf{x}}_{0,1,0}^{(i,j)} \rightarrow \tilde{\mathbf{x}}_{0,1,1}^{(i,j)} \rightarrow \cdots \rightarrow \tilde{\mathbf{x}}_{0,1,n_f}^{(i,j)} \\
&\rightarrow \quad \vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad \vdots \\
&\rightarrow \tilde{\mathbf{x}}_{n_h,n_w,0}^{(i,j)} \rightarrow \tilde{\mathbf{x}}_{n_h,n_w,1}^{(i,j)} \rightarrow \cdots \rightarrow \tilde{\mathbf{x}}_{n_h,n_w,n_f}^{(i,j)}.
\end{aligned} \tag{12}$$

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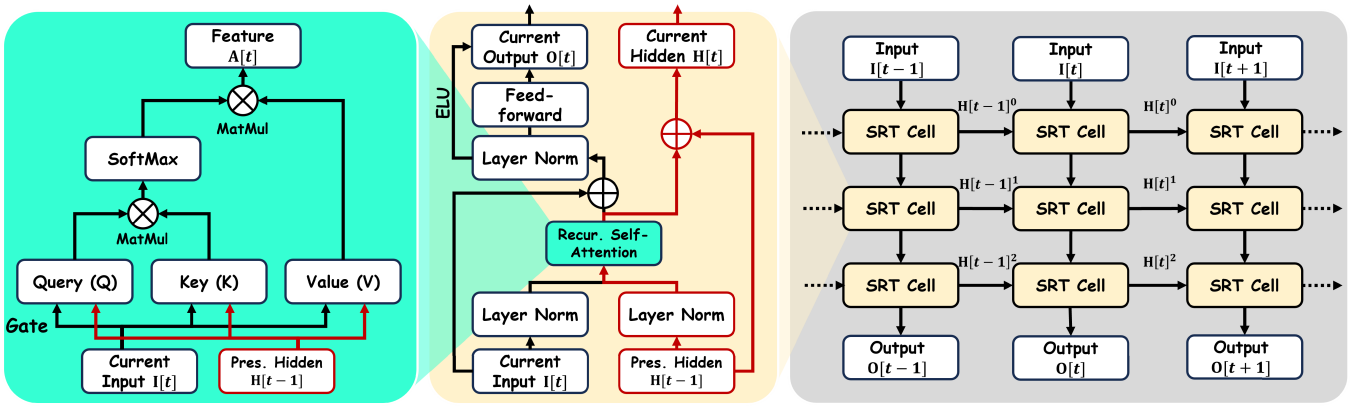


Fig. 7. Illustration of SRT. (a) Recurrent self-attention module focuses on keeping important parts of previous data while adding new information from the latest input. (b) Each SRT cell works with two types of information: current input and previous hidden state, to produce two outputs: the immediate current output and an updated current hidden. (c) This shows an unfolded three-layer structure of the SRT model.

Subsequently, by arranging each subcube into a vector with the size of $\mathbb{R}^{1\times D_c}$, we can obtain the 1th scanned sequence

$$\mathbf{S}_1^{(i,j)} = \Big[\tilde{\mathbf{x}}_{0,0,0}^{(i,j)}, \tilde{\mathbf{x}}_{0,0,1}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{0,0,n_f}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{n_h,n_w,0}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{n_h,n_w,n_f}^{(i,j)}\Big]^\top. \quad (13)$$

Here, $\mathbf{S}_1^{(i,j)}$ belongs to the space $\mathbb{R}^{N_c\times D_c}$ and is referred to as the cubic-sequence. Within this structure, each segment that encompasses a set of subcubes sharing the same spatial coordinates (u, v) constitutes what we define as a "short-term" feature, characterized by spectral continuity. Conversely, the "long-term" feature integrates all "short-term" features, thus encapsulating the spatial contextual information.

Unlike previous works [49], [50], [51], which only scan the HSI patch spatially, this study extends the scanning to a 3-D version that includes band-to-band continuity. Moreover, each element of the scanned target is a unit of a "local" spatial–spectral feature cube instead of a spectral signature. Therefore, the arrangement of scanned features is not merely spectral, but a locally convolved spatial–spectral feature.

E. Cubic-Net Framework

Moving forward, we propose a Cubic-Net to process the cubic-sequence. This approach is designed to capture dependencies both locally (short-term features) and globally (long-term features), allowing for a comprehensive analysis.

1) Cubic-Sequence Segmentation: To elucidate the spectral continuous feature and the spatial contextual feature, the segmentation of the cubic-sequence is performed. This segmentation involves dividing the sequence into multiple subsequences with equal length.

Considering (13), the sequence $\mathbf{S}_1^{(i,j)}$ is segmented into subsequences, each encompassing a length of n_f. These subsequences are tailored to capture the spectral variations across each unique pair of height and width indices within the "hypercube." Thus, each subsequence encapsulates the spectral signature pertinent to a specific spatial coordinate, delineating the continuous spectral feature for in-depth analysis.

We denote each subsequence as $\mathbf{S}_{1,t}^{(i,j)}$, where t represents the subsequence index. The sequence $\mathbf{S}_1^{(i,j)}$ is decomposed into several subsequences, as illustrated in Fig. 6

$$\mathbf{S}_{1,1}^{(i,j)} = \Big[\tilde{\mathbf{x}}_{0,0,0}^{(i,j)}, \tilde{\mathbf{x}}_{0,0,1}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{0,0,n_f}^{(i,j)}\Big]^\top \quad (14a)$$

$$\mathbf{S}_{1,2}^{(i,j)} = \Big[\tilde{\mathbf{x}}_{0,1,0}^{(i,j)}, \tilde{\mathbf{x}}_{0,1,1}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{0,1,n_f}^{(i,j)}\Big]^\top \quad (14b)$$

$$\vdots$$

$$\mathbf{S}_{1,N_s}^{(i,j)} = \Big[\tilde{\mathbf{x}}_{n_h,n_w,0}^{(i,j)}, \tilde{\mathbf{x}}_{n_h,n_w,1}^{(i,j)}, \ldots, \tilde{\mathbf{x}}_{n_h,n_w,n_f}^{(i,j)}\Big]^\top \quad (14c)$$

where $N_s = n_h \times n_w$ denotes the total number of subsequences. Each subsequence $\mathbf{S}_{1,t}^{(i,j)} \in \mathbb{R}^{n_f\times D_c}$ consists of number of n_f elements. Here, t varies from 1 to N_s, representing all combinations of height and width indices.

2) Segmented RT: Numerous studies have validated the effectiveness of incorporating recurrent dynamics (specifically RNNs) into transformer architectures, merging their respective strengths while mitigating their individual limitations to enhance model performance. Nevertheless, the prevalent strategy of integrating RNNs with transformers through a cascade or parallel way often results in substantial computational demands and increased complexity.

Hence, our study introduces the design of the SRT, which is tailored to intrinsically process sequential data by leveraging both an inherent ordering bias and a self-attention mechanism. The intricate architecture of our model is depicted in Fig. 7.

For sequential data denoted by $\mathbf{I} = [\mathbf{I}[0], \mathbf{I}[1], \mathbf{I}[2], \ldots, \mathbf{I}[n]]^\top$ of length n, where each element of $\mathbf{I}$ may be a matrix or vector, the output of the SRT cell, represented as sequential features of the same length, can be computed as follows:

$$\mathbf{O}[t], \mathbf{H}[t] = \phi(\mathbf{I}[t], \mathbf{H}[t-1]). \quad (15)$$

In this equation, $\phi()$ denotes the SRT cell operation, illustrated in Fig. 7(b). $\mathbf{O}[t]$ signifies the output at the current time step t, and $\mathbf{H}[t]$ signifies the hidden feature at time step t. The hidden state $\mathbf{H}[t]$ is initialized with zeros when $t = 0$.

Within the SRT cell, the core component is the recurrent self-attention module, as shown in Fig. 7(a). This module

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integrates the current input $\mathbf{I}[t]$ with the previous hidden state $\mathbf{H}[t-1]$ through self-attention mechanisms. This integration effectively captures the sequential dependencies, marrying the inherent ordering bias with the attention mechanism across the sequence. The recurrent self-attention module enables the model to maintain a continuous state across the sequence, enriching the feature representation with both the current input and the accumulated knowledge from previous states

$$\begin{bmatrix}\mathbf{Q}[t]\\ \mathbf{K}[t]\\ \mathbf{V}[t]\end{bmatrix} = \begin{bmatrix}\alpha^q\left(\mathbf{I}[t]\,\mathbf{W}_I^Q\right)+\beta^q\left(\mathbf{H}[t-1]\,\mathbf{W}_H^Q\right)\\ \alpha^k\left(\mathbf{I}[t]\,\mathbf{W}_I^K\right)+\beta^k\left(\mathbf{H}[t-1]\,\mathbf{W}_H^K\right)\\ \alpha^v\left(\mathbf{I}[t]\,\mathbf{W}_I^V\right)+\beta^v\left(\mathbf{H}[t-1]\,\mathbf{W}_H^V\right)\end{bmatrix} \tag{16}$$

where the variables $\mathbf{Q}[t]$, $\mathbf{K}[t]$, and $\mathbf{V}[t]$ are designated as the current query, key, and value matrices, respectively. In addition, the weights $\mathbf{W}_I^Q$, $\mathbf{W}_I^K$, and $\mathbf{W}_I^V$ correspond to the input transformations for these matrices, while $\mathbf{W}_H^Q$, $\mathbf{W}_H^K$, and $\mathbf{W}_H^V$ are the transformation weights applied to the hidden state. Furthermore, the scalars α and β act as attention gates that dynamically modulate the influence of the current and past states. Notably, β is computed as the complement of α passed through a sigmoid function, i.e., $\beta = 1-\text{Sigmoid}(\alpha)$, enabling the model to adaptively balance between the new input and the historical context at each time step.

Subsequently, $\mathbf{Q}[t], \mathbf{K}[t], and \mathbf{V}[t]$ follows the general self-attention operation to get the attention feature $\mathbf{a}[t]$ as:

$$\mathbf{a}[t] = \text{Softmax}\left(\frac{\mathbf{Q}[t]\,(\mathbf{K}[t])^\top}{\sqrt{d_k}}\right)\mathbf{V}[t] \tag{17}$$

where d_k is the dimension of $\mathbf{K}[t]$.

To leverage the benefits of multihead self-attention, we extend the above formulation by combining the outputs of h independent attention heads

$$\mathbf{A}[t] = \text{Concat}\left(\mathbf{a}[t]^1, \mathbf{a}[t]^2, \mathbf{a}[t]^3, \ldots, \mathbf{a}[t]^h\right)\mathbf{U}_A \tag{18}$$

where each $\mathbf{a}[t]^i$ is the attention output of head i and $\mathbf{U}_A$ is the linear transformation matrix applied to the concatenated attention outputs from all heads.

Next, the current hidden state $\mathbf{H}[t]$ for each time step t is updated by summing the previous hidden state $\mathbf{H}[t-1]$ with the attention features $\mathbf{A}[t]$

$$\mathbf{H}[t] = \mathbf{H}[t-1] + \mathbf{A}[t]. \tag{19}$$

To calculate the output $\mathbf{O}[t]$, a two-step process is applied. First, a candidate $\hat{\mathbf{O}}[t]$ is generated by normalizing the sum of the current input $\mathbf{I}[t]$ and the attention features $\mathbf{A}[t]$

$$\hat{\mathbf{O}}[t] = \text{LN}(\mathbf{I}[t] + \mathbf{A}[t]) \tag{20}$$

where LN() denotes the layer normalization function. Subsequently, the final output $\mathbf{O}[t]$ is calculated by

$$\mathbf{O}[t] = \text{FFN}\left(\hat{\mathbf{O}}[t]\right) + \text{ELU}\left(\hat{\mathbf{O}}[t]\right). \tag{21}$$

Here, FFN() represents the feed-forward network operation, and ELU() is the Exponential Linear Unit function, which is commonly used in neural networks to introduce nonlinearity without reducing the mean activation to zero, thus helping to mitigate the vanishing gradient problem.

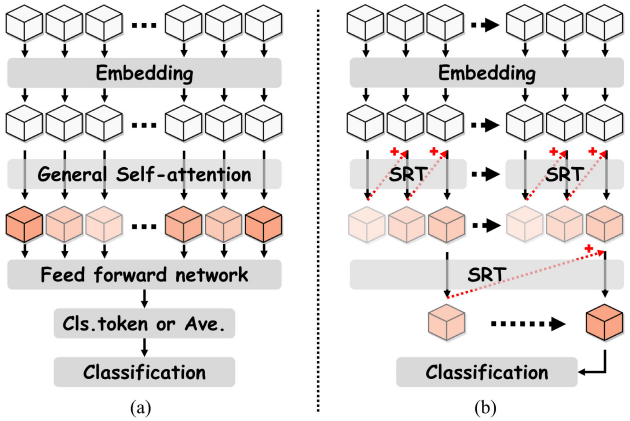


Fig. 8. Conceptual difference between (a) self-attention in original transformer and (b) proposed SRT for handling the cubic-sequence.

Certainly, the SRT cell can be stacked into several layers denoted as "depth" as same as the prior works (i.e., deep RNN, multilayer transformers). Fig. 7(c) shows an illustration of stacking three layers of SRT cell

$$\mathbf{O}[t]^{(l)}, \mathbf{H}[t]^{(l)} = \phi\left(\mathbf{O}[t]^{(l-1)}, \mathbf{H}[t-1]^{(l)}\right) \tag{22}$$

where l denotes the lth layer. $\phi()$ denotes the SRT cell operation mentioned before.

As shown in Fig. 8, unlike the original self-attention mechanism, which processes all elements of the sequence simultaneously and in parallel, our SRT with its segmented recurrent self-attention mechanism tends to handle the sequence locally and recursively. Moreover, compared with methods that simply stack RNNs with Transformers, our proposal inherently designs self-attention to operate in a recurring manner.

3) Short-Term Feature With SRT: Given the presence of continuous spectral features in each subsequence at a specific spatial location, it is logical to conceive a band-to-band ordering in a recurrent manner, as illustrated in Fig. 9(a).

For a given subsequence $\mathbf{S}_{m,t}^{(i,j)} \in \mathbb{R}^{n_f \times D_c}$, where m represents the mth scan order, and t the tth subsequence, we initially convert it into an image-sequence. The transformed size is $\mathbf{S}_{m,t}^{(i,j)} \in \mathbb{R}^{n_f \times (d_c \times p_c^2)}$. Here, n_f denotes the number of images (frames), and $d_c \times p_c^2$ represents the dimensions of each frame. This implies that each element $\tilde{\mathbf{x}}_{u,v,z}^{(i,j)} \in \mathbb{R}^{1 \times D_c}$ within the subsequence is modified into $\tilde{\mathbf{x}}_{u,v,z}^{(i,j)} \in \mathbb{R}^{d_c \times p_c^2}$, adhering to the previously mentioned condition that $D_c = p_c \times p_c \times d_c$.

Hence, given a current zth frame $\mathbf{S}_{m,t}^{(i,j)}[z] \in \mathbb{R}^{d_c \times p_c^2}$ in the subsequence and a hidden state $\mathbf{H}^{\mathbf{st}}[z-1] \in \mathbb{R}^{d_c \times p_c^2}$ from the previous frame as the input, the recurrent self-attention is designed to establish an interaction between them. It generates an attended feature $\mathbf{A}^{\mathbf{st}}[z] \in \mathbb{R}^{d_c \times p_c^2}$ as follows:

$$\mathbf{A}^{\mathbf{st}}[z] = \text{Softmax}\left(\frac{\mathbf{Q}^{\mathbf{st}}[z]\,(\mathbf{K}^{\mathbf{st}}[z])^\top}{\sqrt{d_k}}\right)\mathbf{V}^{\mathbf{st}}[z] \tag{23}$$

where d_k is the feature dimension of $\mathbf{K}^{\mathbf{st}}[z]$, and "**st**" denotes the operation at the short-term level. $\mathbf{Q}^{\mathbf{st}}[z]$, $\mathbf{K}^{\mathbf{st}}[z]$, and $\mathbf{V}^{\mathbf{st}}[z]$ represent the current query, key, and value matrices,

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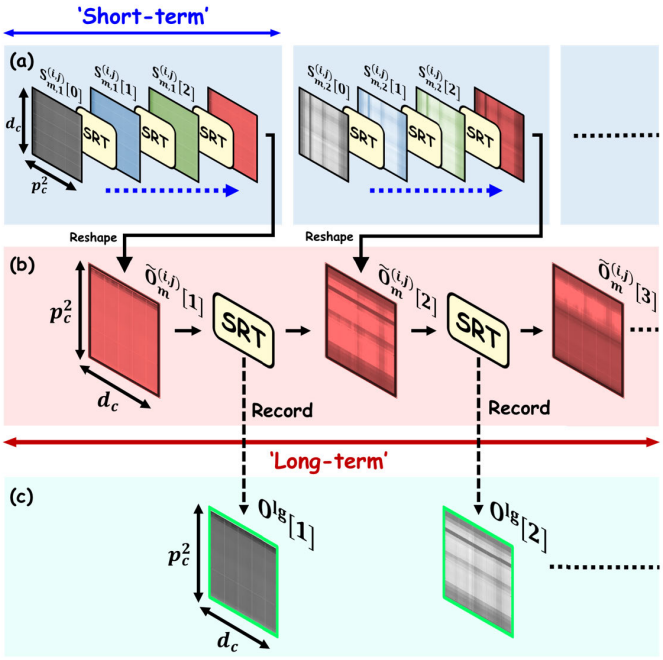


Fig. 9. Illustration of (a) short-term feature processing with SRT, where the input for each short-term module is the subsequence $\mathbf{S}_{m,t}^{(i,j)}$. (b) Long-term feature processing with SRT, using $\tilde{\mathbf{O}}_m^{(i,j)}$ as the input for the long-term module. (c) Feature recording at each step in the process, with each output from the SRT cells in the long-term module being recorded as $\mathbf{O}^{\mathbf{lg}}[t]$ and passed to the subsequent step.

respectively, defined as

$$\begin{bmatrix}\mathbf{Q}^{\mathbf{st}}[z]\\ \mathbf{K}^{\mathbf{st}}[z]\\ \mathbf{V}^{\mathbf{st}}[z]\end{bmatrix} = \begin{bmatrix}\alpha^q\left(\mathbf{S}_{m,t}^{(i,j)}[z]\,\mathbf{W}_S^Q\right)+\beta^q\left(\mathbf{H}^{\mathbf{st}}[z-1]\,\mathbf{W}_H^Q\right)\\ \alpha^k\left(\mathbf{S}_{m,t}^{(i,j)}[z]\,\mathbf{W}_S^K\right)+\beta^k\left(\mathbf{H}^{\mathbf{st}}[z-1]\,\mathbf{W}_H^K\right)\\ \alpha^v\left(\mathbf{S}_{m,t}^{(i,j)}[z]\,\mathbf{W}_S^V\right)+\beta^v\left(\mathbf{H}^{\mathbf{st}}[z-1]\,\mathbf{W}_H^V\right)\end{bmatrix}. \tag{24}$$

Next, the current hidden state $\mathbf{H}^{\mathbf{st}}[z] \in \mathbb{R}^{d_c\times p_c^2}$ can be updated in a residual manner

$$\mathbf{H}^{\mathbf{st}}[z] = \mathbf{H}^{\mathbf{st}}[z-1] + \mathbf{A}^{\mathbf{st}}[z]. \tag{25}$$

Subsequently, the current output $\mathbf{O}^{\mathbf{st}}[z] \in \mathbb{R}^{d_c\times p_c^2}$ from the SRT is computed as follows:

$$\mathbf{O}^{\mathbf{st}}[z] = \mathrm{FFN}\left(\hat{\mathbf{O}}^{st}[z]\right) + \mathrm{ELU}\left(\hat{\mathbf{O}}^{st}[z]\right) \tag{26}$$

$$\hat{\mathbf{O}}^{st}[z] = \mathrm{LN}\left(\mathbf{S}_{m,t}^{(i,j)}[z] + \mathbf{A}^{\mathbf{st}}[z]\right). \tag{27}$$

Consequently, by processing all frames in a subsequence, the final output representation $\mathbf{O}_{m,t}^{(i,j)} \in \mathbb{R}^{d_c\times p_c^2}$ for the subsequence $\mathbf{S}_{m,t}^{(i,j)}$ is defined as

$$\mathbf{O}_{m,t}^{(i,j)} = \mathbf{O}^{\mathbf{st}}[n_f] \tag{28}$$

where n_f denotes the final frame index in a subsequence.

Applying the same operations to all subsequences at the mth scan order, we can construct a new representation as

$$\tilde{\mathbf{O}}_m^{(i,j)} = \mathrm{Concat}\left(\mathbf{O}_{m,1}^{(i,j)}, \mathbf{O}_{m,2}^{(i,j)}, \ldots, \mathbf{O}_{m,N_s}^{(i,j)}\right) \tag{29}$$

where $\tilde{\mathbf{O}}_m^{(i,j)} \in \mathbb{R}^{d_c\times p_c^2\times N_s}$ and N_s represents the total number of subsequences. It's important to note that each output is obtained independently.

Following this, the newly formed representation is prepared for long-term feature aggregation in the next stage.

4) Long-Term Feature With SRT: Given the aforementioned feature $\tilde{\mathbf{O}}_m^{(i,j)}$, which represents the spectral information processed through recurrence, we first transform it to the dimensions of $\mathbb{R}^{p_c^2\times d_c\times N_s}$.

As a result, each element within $\tilde{\mathbf{O}}_m^{(i,j)}[t]$ can be interpreted as a spatial feature corresponding to the tth step, where t ranges from 1 to N_s.

As illustrated in Fig. 9(b), paralleling the preceding process, we consider the current spatial feature $\tilde{\mathbf{O}}_m^{(i,j)}[t]$ along with the preceding spatial step's hidden state $\mathbf{H}^{\mathbf{lg}}[t-1] \in \mathbb{R}^{p_c^2\times d_c}$ as inputs. The recurrent self-attention mechanism is then implemented as follows:

$$\mathbf{A}^{\mathbf{lg}}[t] = \mathrm{Softmax}\left(\frac{\mathbf{Q}^{\mathbf{lg}}[t]\,(\mathbf{K}^{\mathbf{lg}}[t]\,)^\top}{\sqrt{d_k}}\right)\mathbf{V}^{\mathbf{lg}}[t] \tag{30}$$

where d_k represents the feature dimension of $\mathbf{K}^{\mathbf{lg}}[t]$, and "**lg**" indicates operations at the long-term level. $\mathbf{Q}^{\mathbf{lg}}[t]$, $\mathbf{K}^{\mathbf{lg}}[t]$, and $\mathbf{V}^{\mathbf{lg}}[t]$ are defined as the current Query, Key, and Value matrices within the long-term operation

$$\begin{bmatrix}\mathbf{Q}^{\mathbf{lg}}[t]\\ \mathbf{K}^{\mathbf{lg}}[t]\\ \mathbf{V}^{\mathbf{lg}}[t]\end{bmatrix} = \begin{bmatrix}\alpha^q\left(\tilde{\mathbf{O}}_m^{(i,j)}[t]\,\mathbf{U}_S^Q\right)+\beta^q\left(\mathbf{H}^{\mathbf{lg}}[t-1]\,\mathbf{U}_H^Q\right)\\ \alpha^k\left(\tilde{\mathbf{O}}_m^{(i,j)}[t]\,\mathbf{U}_S^K\right)+\beta^k\left(\mathbf{H}^{\mathbf{lg}}[t-1]\,\mathbf{U}_H^K\right)\\ \alpha^v\left(\tilde{\mathbf{O}}_m^{(i,j)}[t]\,\mathbf{U}_S^V\right)+\beta^v\left(\mathbf{H}^{\mathbf{lg}}[t-1]\,\mathbf{U}_H^V\right)\end{bmatrix}. \tag{31}$$

In this scenario, the SRT operates through a tripartite process. Initially, the spatial feature $\tilde{\mathbf{O}}_m^{(i,j)}[t]$ and the preceding hidden state $\mathbf{H}^{\mathbf{lg}}[t-1]$ undergo LN(), subsequently feeding into the recurrent self-attention mechanism. This mechanism adeptly balances the retention of prior attention details while assimilating new insights from the current feature. Concluding this sequence, the updated hidden state $\mathbf{H}^{\mathbf{lg}}[t] \in \mathbb{R}^{p_c^2\times d_c}$ emerges as a result of an additive integration

$$\mathbf{H}^{\mathbf{lg}}[t] = \mathbf{H}^{\mathbf{lg}}[t-1] + \mathbf{A}^{\mathbf{lg}}[t]. \tag{32}$$

This formula delineates how the current hidden state is derived by summing the previous state with the newly computed attention feature $\mathbf{A}^{\mathbf{lg}}[t]$.

The generation of the current output $\mathbf{O}^{\mathbf{lg}}[t] \in \mathbb{R}^{p_c^2\times d_c}$ within the SRT proceeds as follows:

$$\mathbf{O}^{\mathbf{lg}}[t] = \mathrm{FFN}\left(\hat{\mathbf{O}}^{lg}[t]\right) + \mathrm{ELU}\left(\hat{\mathbf{O}}^{lg}[t]\right) \tag{33}$$

where $\hat{\mathbf{O}}^{lg}[t]$ is computed by LN() applied to the combination of the current spatial feature $\tilde{\mathbf{O}}_m^{(i,j)}[t]$ and the current attention feature $\mathbf{A}^{\mathbf{lg}}[t]$

$$\hat{\mathbf{O}}^{lg}[t] = \mathrm{LN}\left(\tilde{\mathbf{O}}_m^{(i,j)}[t] + \mathbf{A}^{\mathbf{lg}}[t]\right). \tag{34}$$

Subsequently, as each spatial step is processed, the outputs, encapsulating the spatial contextual features corresponding to the mth scan order, are aggregated as follows:

$$\mathbf{F}_m^{(i,j)} = \left[\mathbf{O}^{\mathbf{lg}}[1], \mathbf{O}^{\mathbf{lg}}[2], \mathbf{O}^{\mathbf{lg}}[3], \ldots \mathbf{O}^{\mathbf{lg}}[N_s]\right]. \tag{35}$$

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This equation illustrates that $\mathbf{F}_m^{(i,j)}$ captures the output at each spatial step, effectively recording the evolving spatial context as depicted in Fig. 9(c).

F. Multiscanning Feature Fusion

The inherent recurrent design of the SRT, akin to that of RNNs, leads to varying outputs contingent upon the sequence's scan order. This characteristic is pivotal in the context of a 3-D-multiscanning strategy, which encompasses multiple scan orders, each yielding a distinct output representation. Notably, within this strategy, certain scan orders naturally form inverse pairs, for example, the pair comprising scan order-1 and scan order-3. This observation lends itself to the intuitive adoption of a bi-directional approach, similar to prior research utilizing bi-directional RNNs for data processing [73], [74], [75], [76], [77].

1) Bi-Directional Feature Aggregation: As depicted in Fig. 2, the methodology borrows from the design principles of bi-directional RNNs. Upon acquiring two sets of output features from corresponding inverse scan orders, these are seamlessly merged into a singular, comprehensive feature. Specifically, the forward feature, denoted as $\overrightarrow{\mathbf{F}}_m^{(i,j)}$, and the backward feature, represented by $\overleftarrow{\mathbf{F}}_m^{(i,j)}$, are defined as follows:

$$\overrightarrow{\mathbf{F}}_m^{(i,j)} = \left[\overrightarrow{\mathbf{O}^{\mathrm{lg}}}[1], \overrightarrow{\mathbf{O}^{\mathrm{lg}}}[2], \overrightarrow{\mathbf{O}^{\mathrm{lg}}}[3], \ldots, \overrightarrow{\mathbf{O}^{\mathrm{lg}}}[N_s]\right] \tag{36a}$$

$$\overleftarrow{\mathbf{F}}_m^{(i,j)} = \left[\overleftarrow{\mathbf{O}^{\mathrm{lg}}}[1], \overleftarrow{\mathbf{O}^{\mathrm{lg}}}[2], \overleftarrow{\mathbf{O}^{\mathrm{lg}}}[3], \ldots, \overleftarrow{\mathbf{O}^{\mathrm{lg}}}[N_s]\right]. \tag{36b}$$

For instance, in this formulation, $\overrightarrow{\mathbf{F}}_m^{(i,j)}$ and $\overleftarrow{\mathbf{F}}_m^{(i,j)}$ correspond to outputs from $\mathbf{F}_1^{(i,j)}$ and $\mathbf{F}_3^{(i,j)}$, respectively. These are then concatenated to form a unified feature representation

$$\tilde{\mathbf{F}}_e^{(i,j)} = \mathrm{Concat}\left(\overrightarrow{\mathbf{F}}_m^{(i,j)}, \overleftarrow{\mathbf{F}}_m^{(i,j)}\right). \tag{37}$$

This concatenated feature $\tilde{\mathbf{F}}_e^{(i,j)}$ encapsulates a richer, more comprehensive representation by integrating perspectives from both the forward and backward scan orders.

Consequently, with the implementation of this bi-directional approach, we obtain multiple pairs of combined features, which can be formally represented as $\{\tilde{\mathbf{F}}_e^{(i,j)}, e = 1, 2, \ldots, E\}$, where $E = M/2$, and M signifies the total number of scan orders used in the process. In our experimental design, we have set M to 8, resulting in the derivation of four distinct pairs of bi-directional features.

2) Feature-Level Fusion With Multiscanning: To effectively integrate the various pairs of features in the decoder for classification, careful consideration of the fusion strategy is imperative. Merely fusing these features without discriminating their individual contributions could obscure their respective positive or negative influences. In response to it, we draw inspiration from [21] and implement two distinct strategies for assigning attention weights to each pair of scan orders.

In the first strategy, we focus on a feature-level fusion process, moving beyond a simplistic summation approach. This method involves a weighted concatenation of the output features, where each feature set is assigned a distinct weight, reflecting its importance or relevance in the overall context. The computation for the input to the output layer is formulated as

$$\mathbf{Y} = \mathrm{Concat}\left(w_1\tilde{\mathbf{F}}_1^{(i,j)}, w_2\tilde{\mathbf{F}}_2^{(i,j)}, \ldots, w_E\tilde{\mathbf{F}}_E^{(i,j)}\right) \tag{38}$$

where $w_e \in \mathbb{R}^1$ and $e \in \{1, 2, \ldots, E\}$ represent the fusion weights for each feature set $\tilde{\mathbf{F}}_e^{(i,j)}$. These weights are not fixed but are integrated into the overall network architecture, allowing them to be optimized automatically through learning from the data. This adaptive weighting mechanism ensures that each feature set contributes in proportion to its relevance, enabling a more nuanced and effective fusion process.

Following the feature fusion, the weighted feature set $\mathbf{Y}$ is then processed through a decoder, specifically a multiLayer perceptron (MLP), for model training. This step is crucial for refining and interpreting the fused features for the final task

$$\tilde{\mathbf{Y}} = \mathrm{MLP}(\mathbf{Y}). \tag{39}$$

The model training is guided by the cross-entropy loss function, a standard choice for classification tasks. This loss function quantifies the difference between the predicted labels and the actual labels of the training samples. It is defined as

$$\mathcal{L} = -\frac{1}{N_{\mathrm{tr}}}\sum_{\mathrm{tr}=1}^{N_{\mathrm{tr}}}\left[\mathbf{L}_{\mathrm{tr}}\log(\tilde{\mathbf{Y}}_{\mathrm{tr}}) + (1-\mathbf{L}_{\mathrm{tr}})\log(1-\tilde{\mathbf{Y}}_{\mathrm{tr}})\right] \tag{40}$$

where N_{tr} represents the total number of training samples, and $\mathbf{L}_{\mathrm{tr}}$ and $\tilde{\mathbf{Y}}_{\mathrm{tr}}$ denote the true label and the predicted label for the tr-th training sample, respectively. This loss function effectively drives the training process by minimizing the discrepancy between the predicted and true labels.

3) Output-Level Fusion With Multiscanning: In contrast to the first strategy, the second approach employs an output-level connection method. This strategy involves directly feeding the multiscanning features into separate output layers. Each feature from the set $\{\tilde{\mathbf{F}}_e^{(i,j)}, e = 1, 2, \ldots, E\}$ is processed through its own MLP to generate corresponding outputs

$$\tilde{\mathbf{Y}}_e = \mathrm{MLP}\left(\tilde{\mathbf{F}}_e^{(i,j)}\right). \tag{41}$$

From this operation, a collection of loss functions is obtained, denoted as $\{\mathcal{L}_e, e = 1, 2, \ldots, E\}$, following the formulation in (40). To effectively integrate these individual loss functions, a weighted summation method is applied

$$\tilde{\mathcal{L}} = \frac{1}{E}\sum_{e=1}^{E} w_e\mathcal{L}_e. \tag{42}$$

Here, $w_e \in \mathbb{R}^1$ represents the fusion weight for each individual loss function. The final loss function $\tilde{\mathcal{L}}$, which is a weighted combination of the individual loss functions, is then utilized for the overall model training. This method allows for a more nuanced and balanced integration of the various features processed through multiple scan orders.

III. Experimental Results and Analysis

In this section, we first describe three well-known HSI datasets. We then outline the implementation details and the methods used for comparison. Finally, we conduct extensive experiments to evaluate the performance of the proposed method both quantitatively and qualitatively.

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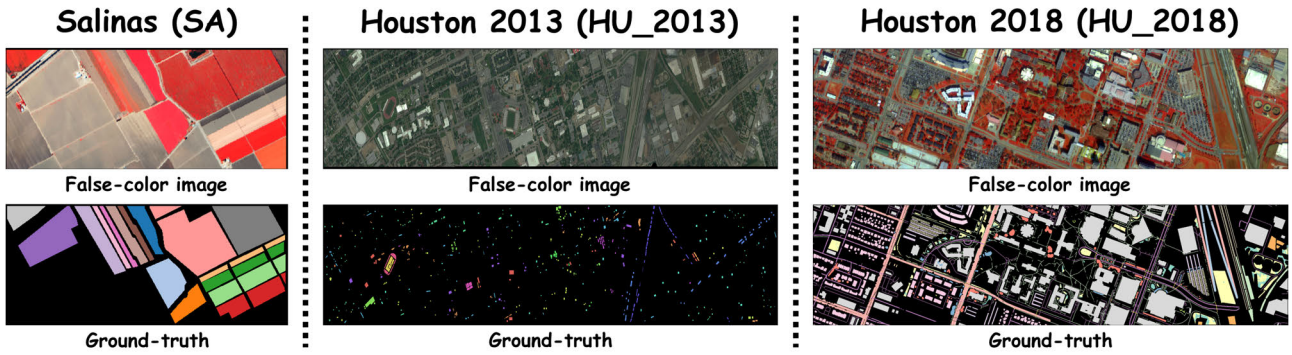


Fig. 10. Three datasets, SA, HU_2013, and HU_2018, are used for experiments. The false-color images and ground-truths are shown above.

TABLE I
DESCRIPTION OF LAND-COVER TYPES AND NUMBER OF TRAIN-TEST SAMPLES FOR THREE DATASETS

	Salinas (SA)				Houston 2013 (HU_2013)				Houston 2018 (HU_2018)			
No.	Class Name	Color	Train	Test	Class Name	Color	Train	Test	Class Name	Color	Train	Test
1	Broccoli-g-w-1		50	1,988	Grass-health		198	1,053	Healthy grass		100	9,699
2	Broccoli-g-w-2		50	3,705	Grass-stress		190	1,064	Stressed grass		100	32,402
3	Fallow		50	1,955	Grass-synthetic		192	505	Artificial turf		100	584
4	Fallow-r-p		50	1,373	Tree		188	1,056	Evergreen trees		100	13,495
5	Fallow-s		50	2,658	Soils		186	1,056	Deciduous trees		100	4,921
6	Stubble		50	3,938	Waters		182	143	Bare earth		100	4,416
7	Celery		50	3,558	Residential		196	1,072	Water		100	166
8	Grapes-u		50	11,250	Commercials		191	1,053	Residential build		100	39,672
9	Soil-v-d		50	6,182	Roads		193	1,059	Non-residential build		100	223,652
10	Corn-s-w		50	3,257	Highways		191	1,036	Roads		100	45,766
11	Lettuce-r-4-w		50	1,047	Railways		181	1,054	Sidewalks		100	33,929
12	Lettuce-r-5-w		50	1,906	Park-l-1		192	1,041	Crosswalks		100	1,418
13	Lettuce-r-6-w		50	895	Park-l-2		184	285	Major thoroughfares		100	46,248
14	Lettuce-r-7-w		50	1,050	Tennis-ground		181	247	Highways		100	9,765
15	Vineyard-u		50	7,247	Running-track		187	473	Railways		100	6,837
16	Vinyard-v-t		50	1,786					Paved parking		100	11,400
17									Unpaved parking		100	36
18									Cars		100	6,447
19									Trains		100	5,269
20									Stadium seats		100	6,524
Total			800	53,329			2,832	12,197			2,000	502,646
Ratio			1.4%	98.6%			18.8%	81.2%			0.4%	99.6%

A. Datasets Description

Three publicly available HSI datasets are illustrated in Fig. 10, along with their corresponding false-color images and ground truths. The interband cross correlation matrices for these datasets are shown in Fig. 11, respectively. The numbers of training and testing samples for each class within these datasets are listed in Table I.

The Salinas (SA) dataset[1] was captured by the AVIRIS sensor over the Salinas Valley. It comprises 204 bands with a resolution of 512 × 217 pixels, encompassing 16 classes with a relatively uniform distribution of ground objects. For the experiments, a small fixed sample of 50 pixels per class is selected for training, while the remainder is used for testing.

The Houston 2013 (HU_2013) dataset[2] was used in the 2013 IEEE GRSS Data Fusion Contest. It features 144 spectral bands and spans an area of 349 × 1905 pixels, covering 15 different land-cover classes. The training and testing samples are selected in a disjoint manner.

Finally, the Houston 2018 (HU_2018) dataset was featured in the 2018 IEEE GRSS Data Fusion Contest.[3] This dataset contains 601 × 2384 pixels across 48 spectral bands and includes 20 land-cover classes. Similar to the SA dataset, a small fixed sample of 100 pixels per class is chosen for training purposes in the experiments, with the rest allocated for testing.

B. Experimental Setting

1) General Setting: For the implementation of our proposal, detailed definitions of the notations and hyperparameter settings are provided in Table II and Algorithm 1. All experiments are conducted using Pytorch 1.8 with a Geforce RTX 3060 GPU and Ubuntu 20.04 system. The code for this work will be made available at our GitHub repository.[4]

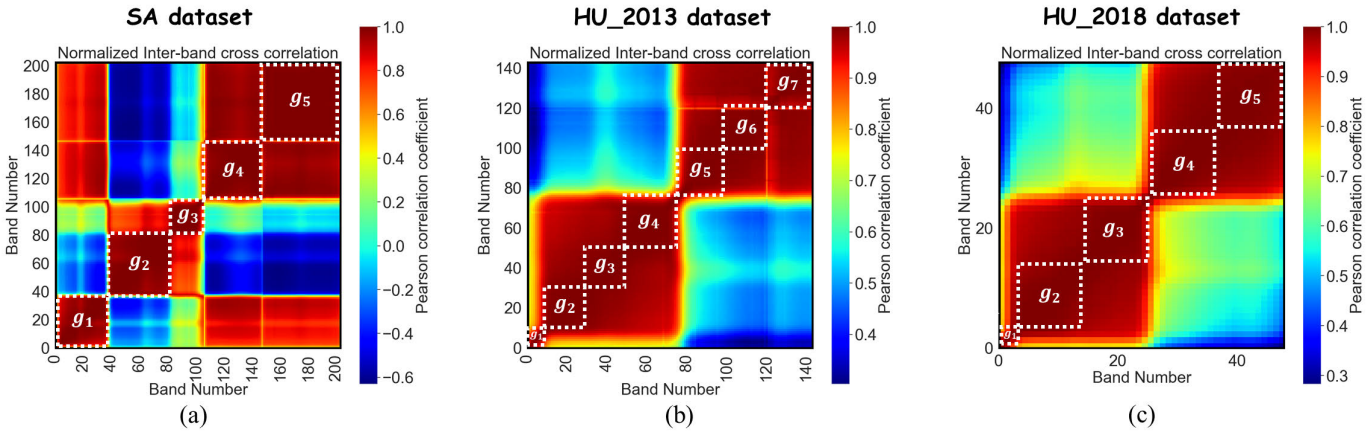
2) Evaluation Indicators: To quantitatively assess the effectiveness of our proposed method and compare it with other methods, we rely on four key performance indicators: overall accuracy (OA), average accuracy (AA), the kappa coefficient

[1]http://lesun.weebly.com/hyperspectral-data-set.html
[2]http://www.grss-ieee.org/community/technical-committees/data-fusion
[3]https://hyperspectral.ee.uh.edu/?page_id=1075
[4]https://github.com/zhouweilian1904/

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TABLE II
HYPER-PARAMETER SETTINGS AND SIZE OF NOTATIONS FOR THREE DATASETS

Definition	Notation	SA	HU_2013	HU_2018
Image size:	$H \times W \times C$	$512 \times 217 \times 204$	$349 \times 1905 \times 144$	$601 \times 2384 \times 48$
Number of Sub-Band Groups:	S	5	7	5
Patch Size:	$p \times p$	15×15	9×9	9×9
Feature Dimension (Output Channel of CNNs):	d	32	16	8
CNN's Kernel Size:	$k_1 \times k_2$	3×3	3×3	3×3
Sub-Cube Size:	$p_c \times p_c \times d_c$	$5 \times 5 \times 8$	$3 \times 3 \times 4$	$3 \times 3 \times 2$
Number of Sub-Cubes:	N_c	180	252	180
Number of Sub-Sequences:	N_s	9	9	9
Length of Each Sub-Sequence:	n_f	20	28	20
Number of short-term SRT layers:	l^{st}	2	2	2
Number of long-term SRT layers:	l^{lg}	2	4	4
Number of multi-heads in short-term SRT:	h^{st}	2	3	3
Number of multi-heads in long-term SRT:	h^{lg}	3	3	2



(a) (b) (c)

Fig. 11. Interband cross correlation results with (a) SA dataset. Five unequal subband groups: $g_1 = [1, 40]$, $g_2 = [41, 80]$, $g_3 = [81, 105]$, $g_4 = [106, 145]$, and $g_5 = [146, 204]$. (b) HU_2013 dataset. Seven unequal subband groups: $g_1 = [1, 10]$, $g_2 = [11, 33]$, $g_3 = [34, 56]$, $g_4 = [57, 80]$, $g_5 = [81, 101]$, $g_6 = [102, 122]$, and $g_7 = [123, 144]$. (c) HU_2018 dataset. Five unequal subband groups: $g_1 = [1, 3]$, $g_2 = [4, 14]$, $g_3 = [15, 25]$, $g_4 = [26, 36]$, and $g_5 = [37, 48]$. It should be noted that the matrices are calculated using MATLAB prior to the neural network modeling. Each HSI dataset has its unique one.

(kappa), and accuracy within each class. These indicators provide a comprehensive view of the classification performance, with higher values indicating superior classification results.

C. Brief Description of Compared Methods

To evaluate the effectiveness of our proposed method, we conducted comparative analyses against a range of baseline and state-of-the-art models. These included General ViT [78], SpeFormer [24], HiT [27], SSFTT [79], 3D-ViT [80], CasRNN [21], Multi-LSTM [50], and Multi-RT [51]. All models were employed as patch-wise classifiers, with parameter settings adhering to their respective references. The comparison is grounded in the same number of training samples.

1) General ViT model, adapted from the original ViT [78], is configured with a feature dimension of 64 and incorporates four multiheads attention. The network depth is set to four layers. Across all datasets, the initial size of the cropped patch is 9×9. We use a subpatch size of 3×3, partitioned in a nonoverlapping manner.
2) SpeFormer integrates group-wise spectral embedding into the transformer encoder and includes cross-layer adaptive fusion modules to boost the encoder's learning capabilities. It is configured with an initial patch size of 7×7 across all datasets in our experiments.
3) HiT embeds convolution operations into the transformer structure to capture the subtle spectral discrepancies and convey the local spatial context information. For all datasets, the initial patch size is set at 15×15 for HiT.
4) SSFTT combines 3D-CNNs and 2D-CNNs, featuring a Gaussian-weighted feature tokenizer to prepare extracted features for the Transformer encoder. In our experiments, patch sizes vary by dataset: 13×13 for the SA dataset and 9×9 for both HU_2013 and HU_2018 datasets.
5) Three-dimensional-ViT integrates spectral and spatial features into subcubes using a 3-D configuration within the ViT framework. It implements 3-D coordinate positional embedding and a local-global feature fusion mechanism, merging local details with global context. For all datasets, the initial patch size for the 3D-ViT model is set at 9×9.
6) CasRNN, processed CNNs and RNNs both alongside HSI's spectral domain with equal subband groups. Two kinds of RNNs are deployed to eliminate spectral redundancy and enhance nonadjacent information. The initial patch size is set as 27×27 for all datasets.
7) Multi-LSTM uses a multiscanning strategy to scan HSI patches into pixel sequences with eight different scan orders, enhancing local patches with correlative

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TABLE III

QUANTITATIVE PERFORMANCE (%) OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE **SALINAS** DATASET. THE BEST ONE IS HIGHLIGHTED

	General ViT [78]	SpeFormer [24]	HiT [27]	SSFTT [79]	3D-ViT [80]	CasRNN [21]	Multi-LSTM [50]	Multi-RT [51]	Ours
1	93.053	77.379	87.086	90.894	**95.673**	92.737	89.401	94.136	94.371
2	96.150	84.375	95.687	96.775	**100**	89.257	97.781	**100**	**100**
3	47.294	82.851	95.905	92.198	89.963	82.399	96.932	97.330	**99.253**
4	94.950	97.512	94.972	96.821	95.770	95.338	93.851	91.397	**97.640**
5	86.920	96.436	95.327	94.288	93.084	81.114	94.682	94.953	**98.667**
6	98.272	98.800	97.502	**98.831**	98.588	98.105	97.570	97.437	97.635
7	93.493	97.719	97.712	**98.921**	98.617	97.591	96.981	97.091	97.966
8	72.057	51.244	80.945	85.917	78.857	80.517	76.497	90.327	**90.992**
9	13.456	99.133	98.963	99.090	**99.138**	90.373	99.124	98.641	98.835
10	64.719	86.627	**94.604**	92.819	83.011	85.116	92.253	93.489	93.362
11	29.499	84.557	94.500	94.993	77.604	41.467	96.103	95.559	**96.494**
12	82.076	85.895	95.707	97.077	**97.283**	59.426	95.057	95.929	96.527
13	74.935	94.276	96.782	**97.722**	96.959	95.259	94.651	96.077	95.583
14	82.580	95.200	95.888	94.192	95.499	94.332	93.373	95.058	**95.574**
15	36.186	64.627	76.211	73.416	52.184	50.264	72.240	81.456	**85.948**
16	86.840	83.425	79.368	87.602	87.184	**91.217**	68.401	73.086	79.134
OA	68.128	79.365	87.183	89.688	85.330	81.401	85.488	90.151	**91.455**
AA	72.030	86.254	92.322	93.222	89.963	83.344	90.931	93.248	**94.874**
Kappa	0.6478	0.7736	0.8587	0.8852	0.8366	0.7928	0.8401	0.8908	**0.9055**

dependence. Concatenated features from these scans are then processed by an RNN for further complementarity. Across all datasets, the initial patch size is set to 7×7.

8) Multi-RT introduces an RT encoder by combining RNNs (specifically, LSTMs) with the ViT into a single unit. It features a multiscanning-controlled positional embedding strategy to enhance feature fusion in the RT encoder. For the SA dataset, the initial patch size is 11×11, while it is 9×9 for both HU_2013 and HU_2018 datasets.

D. Quantitative Results and Classification Maps

Tables III–V show detailed classification results like OA, AA, Kappa, and class-wise accuracy for the SA, HU_2013, and HU_2018 datasets. The classification maps for these datasets are shown in Figs. 12–14.

The methods using only the transformer for classifying HSI are not very effective. General ViT, specifically, does not perform well in classifying the overall images. SpeFormer, while better than the General ViT, still has issues. It often misclassifies areas, creating noise-like patterns in the maps. When looking at the classification maps, it's clear that some areas that should be uniform are not smooth.

Some methods use a mix of CNNs and transformers, such as HiT. While these methods show better results, they have their own limitations. SSFTT combines 3D-CNN, 2D-CNN, and transformers, showing promising outcomes with smoother regions and clearer boundaries. However, the use of deep 3D-CNN and 2D-CNN increases the computational burden.

CasRNN and Multi-LSTM, both RNN-based methods, offer unique approaches to HSI classification. CasRNN combines CNNs with RNNs, focusing on enhancing spectral features. It does this by dividing the HSI into smaller subimages along the spectral domain, concentrating on spectral features. This leads to refined feature representation that reduces redundancies and incorporates nonadjacent information for classification. However, the uniform division on the spectral domain may result in unsatisfctory results. In contrast, Multi-LSTM uses a multiscanning technique to convert HSI patches into multiple-pixel sequences. This method demonstrates the effectiveness of varied sequencing in making RNNs more discerning. As a result, Multi-LSTM outperforms CasRNN, delivering better performance.

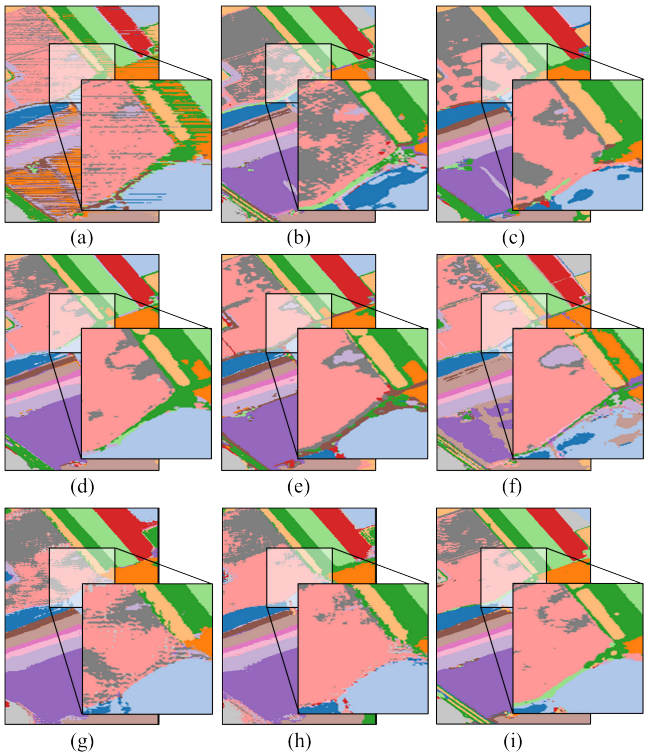


Fig. 12. Classification maps with different methods for the SA dataset. (a) General ViT (OA = 68.128%). (b) SpeFormer (OA = 79.365%). (c) HiT (OA = 87.183%). (d) SSFTT (OA = 89.688%). (e) Three-dimensional-ViT (OA = 85.330%). (f) CasRNN (OA = 81.401%). (g) Multi-LSTM (OA = 85.488%). (h) Multi-RT (OA = 90.151%). (i) Ours (OA = 91.455%).

Three-dimensional-ViT, which processes local spectral and spatial features simultaneously, showcases impressive results with its 3-D concept applied to the ViT model. While not the best performer, it confirms the potential of 3-D model

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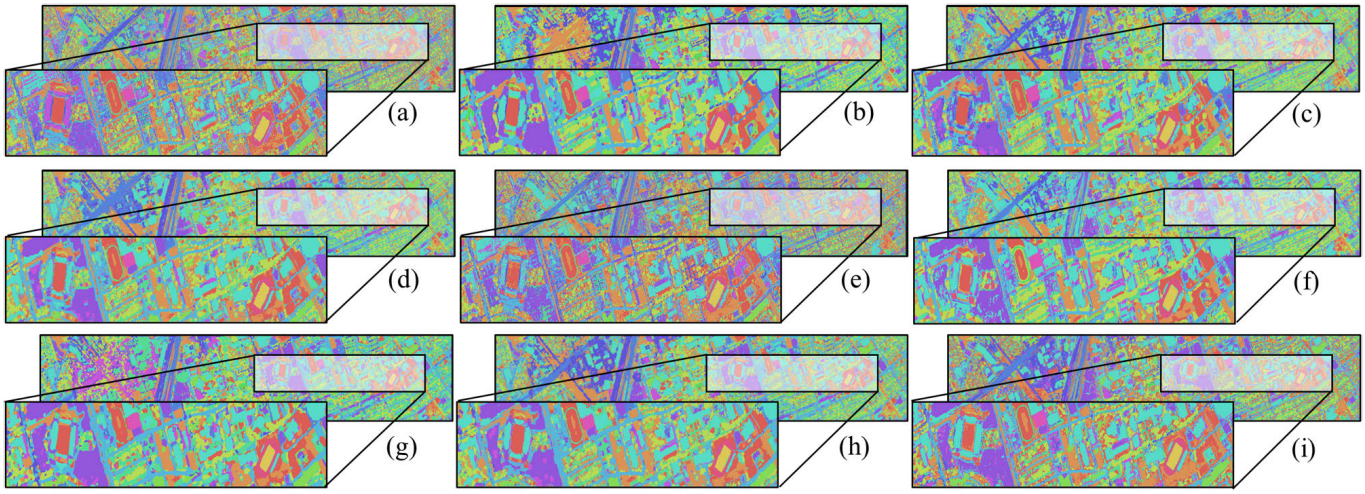


Fig. 13. Classification maps with different methods for the Houston 2013 dataset. (a) General ViT (OA = 79.257%). (b) SpeFormer (OA = 87.235%). (c) HiT (OA = 90.358%). (d) SSFTT (OA = 91.104%). (e) Three-dimensional-ViT (OA = 88.172%). (f) CasRNN (OA = 84.299%). (g) Multi-LSTM (OA = 88.538%). (h) Multi-RT (OA = 92.916%). (i) Ours (OA = 93.884%).

TABLE IV

QUANTITATIVE PERFORMANCE (%) OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE **HOUSTON 2013** DATASET. THE BEST ONE IS HIGHLIGHTED

	General ViT [78]	SpeFormer [24]	HiT [27]	SSFTT [79]	3D-ViT [80]	CasRNN [21]	Multi-LSTM [50]	Multi-RT [51]	Ours
1	89.691	90.112	85.301	90.602	91.213	89.611	90.512	90.410	**92.803**
2	93.872	77.823	94.012	91.694	93.421	92.502	**95.682**	93.601	94.894
3	99.804	99.881	**100**	99.902	96.802	97.921	98.624	**100**	**100**
4	93.625	90.735	91.782	95.584	93.611	96.212	89.203	96.192	**99.640**
5	95.814	99.514	98.354	99.330	97.630	97.734	99.484	99.321	**100**
6	87.182	97.602	97.313	94.483	97.621	97.592	**100**	97.903	**100**
7	80.287	85.333	92.882	87.461	87.323	87.810	77.431	91.224	**93.056**
8	55.060	67.694	88.494	80.222	81.603	73.717	81.522	89.182	**92.691**
9	75.567	91.800	89.612	91.665	74.216	78.192	91.143	**92.214**	87.170
10	61.610	77.772	81.447	87.582	87.281	64.602	77.564	87.232	**88.570**
11	65.223	84.315	86.322	88.504	81.551	95.284	87.622	90.372	**95.468**
12	69.354	94.021	85.353	89.881	70.431	76.793	93.784	**94.093**	89.101
13	41.141	79.723	89.700	89.340	53.934	85.333	60.101	90.714	**93.334**
14	88.363	96.092	**100**	99.201	96.005	95.890	**100**	**100**	**100**
15	99.152	**99.784**	97.564	99.583	98.883	97.755	98.662	100	95.450
OA	79.257	87.235	90.358	91.104	88.172	84.299	88.538	92.916	**93.884**
AA	79.716	88.813	91.876	92.336	86.768	88.463	89.422	94.164	**94.812**
Kappa	0.7752	0.8615	0.8954	0.9035	0.8758	0.8295	0.8758	0.9231	**0.9328**

configurations in ViT applications. In 3D-ViT, both spectral and spatial partitions are uniformly sized, suggesting room for further improvement.

Multi-RT, combining RNNs with transformers with a 2-D-multiscanning strategy, explores the integration of these technologies and benefits from the multiscanning approach, which enhances the sequential model's capability. However, its 2-D-multiscanning strategy is limited to the spatial domain and overlooks the spectral continuity, leading to reduced discriminative power. As a state-of-the-art method, Multi-RT requires additional enhancements to improve performance.

Our study introduces a 3-D-multiscanning-based SRT that recurrences features in both spectral and spatial domains. Compared to Multi-LSTM and Multi-RT, extending scanning to the spectral domain significantly boosts model performance, with a clear improvement in classification accuracy. This highlights the importance of spectral dependency in HSI analysis. Our method's classification maps show smoother regions and clearer land-cover boundaries. By dividing the spectral domain unequally, our approach achieves better class-wise accuracies than CasRNN, SpeFormer, and 3D-ViT.

Among the various methods evaluated, there's a continuing research focus on optimizing spatial–spectral features, particularly through the integration of CNNs, RNNs, and transformers, indicating a significant opportunity for further research and innovation in this domain.

E. Other Analysis and Ablation Study

In this section, a comprehensive analysis, experimental evaluation, and ablation studies have been presented.

1) Selection of Scan Order: In 3-D scanning, the choice of scan order is crucial for feature allocation, including the positioning for sequential models. Thus, determining a suitable scan order across spectral and spatial domains is necessary.

Given that random and casual scanning may lead to significant redundancy and poor feature arrangement, we established two certain rules as follows.

1) Spectrally, scanning should preserve the integrity of band-to-band features.
2) Spatially, scanning should be continuous possibly.

Through investigation, we identified several scanning types, as shown in Fig. 15(a). This included two spectral scanning

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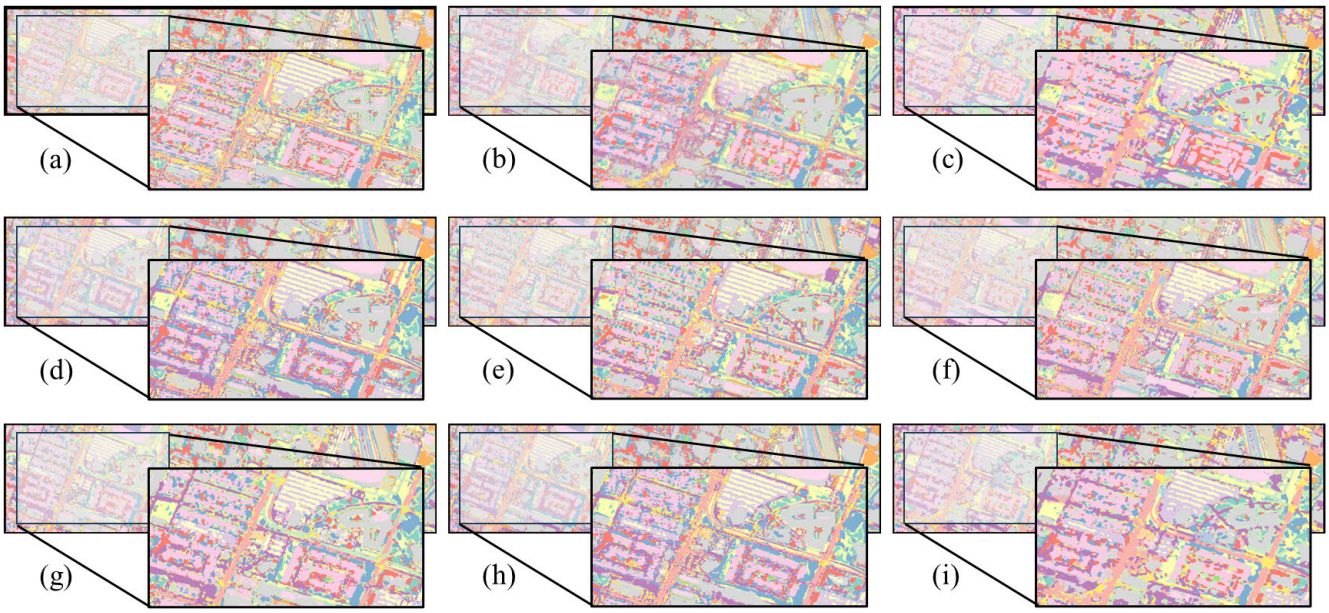


Fig. 14. Classification maps with different methods for the Houston 2018 dataset. (a) General ViT (OA = 57.904%). (b) SpeFormer (OA = 67.968%). (c) HiT (OA = 72.108%). (d) SSFTT (OA = 76.906%). (e) Three-dimensional-ViT (OA = 73.765%). (f) CasRNN (OA = 71.603%). (g) Multi-LSTM (OA = 72.894%). (h) Multi-RT (OA = 76.509%). (i) Ours (OA = 78.947%).

TABLE V

QUANTITATIVE PERFORMANCE (%) OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE **HOUSTON 2018** DATASET. THE BEST ONE IS HIGHLIGHTED

	General ViT [78]	SpeFormer [24]	HiT [27]	SSFTT [79]	3D-ViT [80]	CasRNN [21]	Multi-LSTM [50]	Multi-RT [51]	Ours
1	75.563	55.692	72.070	**85.640**	83.154	55.597	75.307	78.129	75.381
2	84.494	76.538	82.619	**92.396**	89.239	85.910	86.034	85.568	86.722
3	47.255	69.592	78.836	**87.359**	78.367	58.700	63.615	68.104	74.640
4	87.588	77.786	77.415	88.708	91.180	90.004	86.990	**91.521**	88.393
5	35.690	48.017	62.317	66.104	63.947	56.253	58.821	46.887	**67.732**
6	71.925	81.226	83.555	89.751	76.485	88.421	91.573	91.226	**92.558**
7	20.256	69.510	70.226	**71.602**	59.246	78.302	16.179	28.425	44.149
8	70.312	67.158	64.568	74.720	75.243	67.389	75.332	79.560	**82.153**
9	65.189	86.403	87.312	88.336	83.934	83.595	82.294	89.247	**90.666**
10	40.343	43.913	50.153	51.534	51.955	49.835	59.338	55.671	**59.621**
11	42.265	35.924	43.972	54.354	53.745	51.571	54.892	49.536	**56.513**
12	5.657	7.578	8.497	**14.284**	14.243	11.474	13.027	11.559	12.315
13	51.712	42.623	58.194	68.080	66.610	63.614	69.268	65.222	**71.429**
14	75.962	60.685	68.226	83.082	73.796	63.443	**87.786**	82.503	67.740
15	64.132	79.619	92.831	**97.506**	97.038	93.408	92.891	94.550	96.769
16	62.582	72.922	78.922	77.126	78.319	76.579	81.715	83.622	**85.196**
17	2.696	79.646	**93.659**	70.148	8.550	15.817	15.033	32.168	46.231
18	55.706	61.051	**76.295**	49.464	55.000	61.882	54.555	75.354	75.696
19	85.612	77.419	89.918	**94.055**	84.163	82.187	83.673	92.916	76.158
20	34.376	60.900	66.541	81.357	58.186	61.396	46.478	72.577	**90.765**
OA	57.904	67.968	72.108	76.906	73.765	71.603	72.894	76.509	**78.947**
AA	53.996	62.710	70.306	**74.280**	67.120	64.769	64.740	68.717	72.041
Kappa	0.5108	0.6015	0.6548	0.7098	0.6787	0.6523	0.8758	0.7066	**0.7363**

types (straight and snake) and six spatial scanning types (U-turn and raster). In the experiments, each type was paired with a different scanning type from the other domain, creating 12 combinations such as straight-U turn, snake-U turn, straight-raster, and snake-raster. The experiments assessed the performance of each scanning combination. It is important to note that we extended spatial scanning to a bi-directional approach. The results are presented in Fig. 15(b).

The experimental results indicated that straight spectral scanning consistently outperformed snake spectral scanning, regardless of its combination with spatial scanning types. This is attributed to the spectral consistency maintained by independent sequential processing (i.e., short-term SRT). Among spatial scanning patterns, those including jump connections generally yielded lower accuracy. However, U-turn and Hilbert patterns performed better, largely due to their effectiveness in maintaining spatial continuity. Notably, the U-turn pattern showed slightly better performance than the Hilbert pattern. Consequently, after careful consideration, we decided to employ straight spectral scanning combined with U-turn spatial scanning for 3-D continuous cubic scanning.

2) Analysis on Proposed Different Components: In this study, we propose three main components: subband grouping (subband), group-wise convolution (Grp-Conv), and the SRT. To evaluate the efficacy of each component, we conducted an ablation study. The results are detailed in Table VI. It should be noted that when a component was not selected, we implemented an alternative, such as PCA for the subband, general convolution for Grp-Conv, and the RT [51] for SRT.

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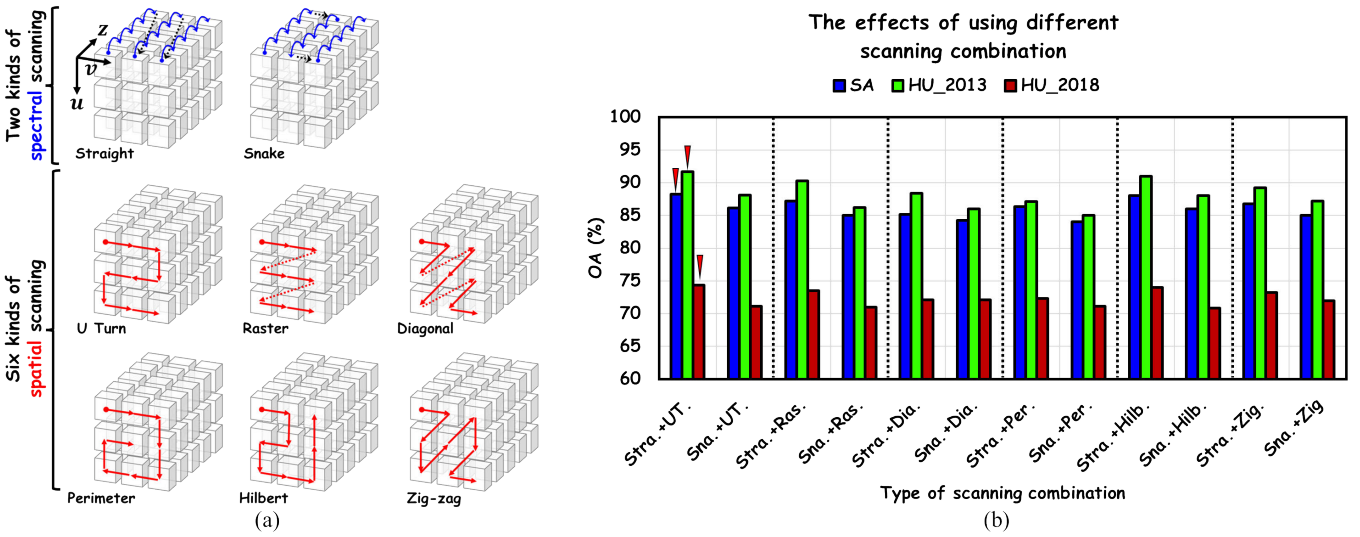


Fig. 15. (a) Investigated scanning types in both spectral and spatial domains. (b) Experimental results are derived from various combinations of spectral and spatial scanning types.

TABLE VI
IMPACT OF DIFFERENT COMPONENTS ON THE CLASSIFICATION RESULTS

Sub-Band.	Grp-Conv.	SRT.	SA	HU_2013	HU_2018
✓			86.158	88.261	75.222
	✓		83.001	85.121	71.555
		✓	83.129	85.888	71.458
✓	✓		90.889	90.886	76.239
✓		✓	90.126	91.214	75.158
	✓	✓	89.999	90.159	75.123
✓	✓	✓	**91.455**	**93.884**	**78.947**

The results indicate that subband grouping and group-wise convolution are most effective when used in conjunction. Using subband alone yields better outcomes than using Grp-Conv by itself. Furthermore, when combined with SRT, the subband exhibits more potential. The usage of SRT demonstrates superior performance compared to its alternate (RT).

Overall, the results lead us to conclude that the three proposed components synergize to enhance performance. The ablation study provides insight into the appropriate handling of image band features with a sequential model. We hold that it is especially important to the sequential analysis with the spectral continuous features in our proposed "short-term" SRT.

3) Analysis on Initial Patch Size: We explore the impact of the initial patch size on classification performance. Two conditions are constrained as follows.

1) The patch size must be an odd number.
2) The patch size must allow for equal slicing in a nonoverlapping manner.

Therefore, we have tested various patch sizes (i.e., 3, 9, 15, 21, and 27) with corresponding spatial sizes of subcubes in our experiments. The results, in terms of OA, for three datasets are presented in Table VII.

Within an optimal range, increasing the initial patch size enhances model performance by incorporating more spatial information. However, larger initial patch sizes can negatively impact results by including more interfering pixels.

For the SA dataset, which is relatively uniform, the optimal HSI patch size and the spatial size of the subcube are verified as 15 × 15 and 5 × 5, respectively. In this case, the optimal number of subsequences is found to be 9. Similarly, for the more complex HU_2013 and HU_2018 datasets with compact spatial distribution, the initial HSI patch size is selected as relatively small, 9 × 9, and the spatial size of the subcube is determined to be 3 × 3. The optimal number of subsequences is also found to be 9.

Furthermore, when the spatial size of subcubes is 1 × 1, it represents the implementation of scanning pixel by pixel. However, this generally results in worse outcomes due to the lack of connections with neighboring pixels. Therefore, as a summary, we decided to implement the 3D partition for the HSI patch for cubic-sequence generation, where each subcube is a unit of 'local' spatial–spectral information.

4) Analysis on Multilayers and Multiheads of SRT: We conducted ablation studies to investigate the impact of varying the number of layers and heads in both short-term and long-term SRTs on their classification performance across three datasets.

For the analysis of multilayers, the results, illustrated in Fig. 16, reveal a performance saturation point; as we increase the depth of both short-term and long-term SRT layers, the OA improves up to an optimal point. Beyond this point, we observe a decline in accuracy, particularly when increasing the depth of short-term SRT layers, a trend that is more pronounced in the HU_2013 and HU_2018 datasets. The optimal depth for short-term and long-term SRT layers is found to be 2, 2, 2, 4, 2, and 4, across the three datasets, respectively. These observations underscore the importance of judiciously selecting hyper-parameters, such as the number of layers, to balance the complexity and generalization capability of the model.

For the analysis of multiheads, the impact of the number of heads in the multihead self-attention mechanism of SRT on classification performance was assessed through experiments, with results focusing on OA depicted in Fig. 17. Increasing

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TABLE VII

IMPACT OF DIFFERENT INITIAL PATCH SIZES AND SPATIAL SIZES OF THE SUBCUBES FOR THE OA (%) ON THREE DATASETS

Initial HSI Patch Size	3×3	9×9		15×15			21×21			27×27		
Spatial Size of Sub-Cubes	1×1	1×1	3×3	1×1	3×3	5×5	1×1	3×3	7×7	1×1	3×3	9×9
SA	83.121	87.321	91.001	85.211	90.129	**91.455**	83.222	87.124	88.345	81.021	85.126	87.009
HU_2013	85.999	91.123	**93.884**	90.012	91.999	92.058	87.981	89.021	91.183	84.321	85.716	88.354
HU_2018	71.002	76.049	**78.947**	75.211	77.088	78.111	72.156	74.125	75.012	66.125	71.056	72.448

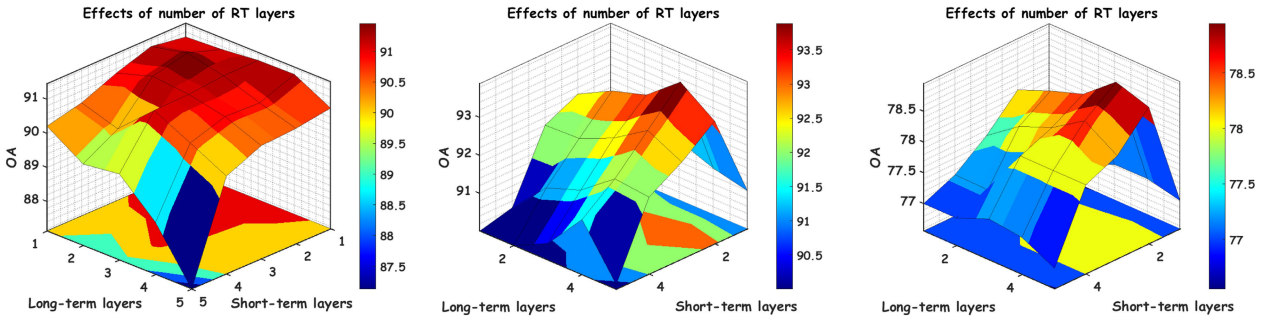


Fig. 16. Effects of depth in short-term SRT and long-term SRT for OA. (a) SA dataset. (b) HU_2013 dataset. (c) HU_2018 dataset. The OA becomes better when increasing the depth of SRTs until a shallow saturation. After that, the OA decreases gradually.

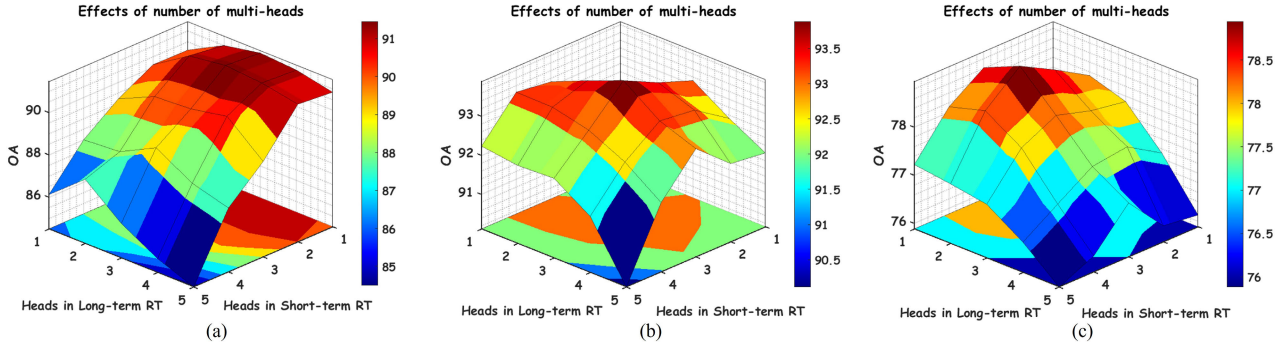


Fig. 17. Effects of multiheads in short-term SRT and long-term SRT for OA. (a) SA dataset. (b) HU_2013 dataset. (c) HU_2018 dataset. The OA becomes better when increasing the number of heads in both SRTs until a shallow saturation. After that, the OA decreases gradually.

the number of heads leads to a modest improvement in OA, with the optimal number of heads varying across datasets. For the SA dataset, the best performance was achieved with two heads in the short-term SRT and three heads in the long-term SRT. For the HU_2013 dataset, both short-term and long-term SRTs achieved optimal results with three heads. In the HU_2018 dataset, the optimal configuration was three heads for short-term SRT and two heads for long-term SRT.

These findings highlight the sensitivity of the model, particularly the short-term SRT, to the number of heads in the multihead self-attention mechanism. Therefore, carefully selecting the number of heads is crucial for optimizing the model's performance, underscoring the importance of this hyper-parameter in the design and tuning of SRT-based systems for classification tasks.

5) Analysis on Number of Subband Groups: To determine the optimal number of subband groups for grouping, we conducted experiments to investigate.

For the SA dataset, the interband cross correlation matrix clearly indicates suitable boundaries for degrouping. However, for the HU_2013 and HU_2018 datasets, the correlation matrices do not provide clear division boundaries due to highly uniform results, as shown in Fig. 11. In these cases, we propose averaging and nonoverlapping division of the highly uniform ranges until a clear division edge is evident. We experimented with averaging 1, 2, 3, 4, and 5 groups in the uniform ranges, with results listed in Tables VIII and IX.

TABLE VIII

IMPACT OF DIFFERENT NUMBER OF AVERAGED GROUPS IN THE UNIFORMED CORRELATION RANGE ON **HU_2013**

	HU_2013: bands' range [11, 80] and [81, 144]				
Averaged Groups	1	2	3	4	5
OA	90.120	92.228	**93.884**	93.312	92.381

Our findings indicate that for the HU_2013 dataset, slicing the uniform range bands [11, 80] and [81, 144] into three averaged groups each yields the best results, resulting in a total of seven subgroups (including area bands [1, 10]). For the HU_2018 dataset, averaging the uniform range bands [4, 25] and [26, 48] into two parts each provides better results, resulting in a total of five subgroups (including area bands [1, 3]).

Although we applied average division to the uniform areas, the start and end band indices for each subgroup are clearly

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TABLE IX

IMPACT OF DIFFERENT NUMBER OF AVERAGED GROUPS IN THE UNIFORMED CORRELATION RANGE ON **HU_2018**

	HU_2018: bands' range [4, 25] and [26, 48]				
Averaged Groups	1	2	3	4	5
OA	74.121	**78.947**	78.411	77.989	77.453

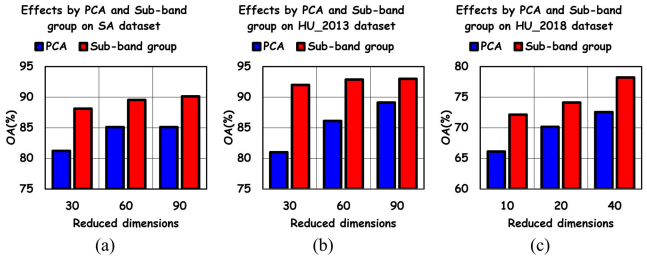


Fig. 18. Comparison of using PCA and proposed subband grouping based group-wise convolution for OA. Experimental results on (a) SA dataset, (b) HU_2013 dataset, and (c) HU_2018 dataset.

visualized. This is a key difference from other end-to-end grouping methods that may not take these into account.

Given that subband grouping is currently performed manually before the end-to-end neural network, it is imperative to develop an automatic version of it for future applications.

6) Effects of Subband Grouping Based Group-Wise Convolution: This section aimed to validate the effectiveness of a subband grouping strategy with group-wise convolution in comparison to using PCA for dimension reduction. This comparison is critical as the proposed SRT model relies on retaining spectral information, making it unsuitable to excessively reduce spectral dimensions. Therefore, for the SA and HU_2013 datasets, the dimensions were selectively set to 30, 60, and 90. For the HU_2018 dataset, which has a shallower original depth, 10, 20, and 40 channels were chosen. The results of these experiments are illustrated in Fig. 18.

The findings demonstrate that the proposed subband grouping strategy consistently outperforms the PCA approach. This indicates two key points: first, that spectral sequential information is valuable for training models like the proposed SRT; and second, PCA may disrupt this sequential continuity.

Therefore, the comparison confirms that subband grouping with group-wise convolution is a more effective method for handling spectral features, particularly when applying sequential models for subsequent processing. This conclusion aligns with findings from previous research [81], [82], [83], which suggest that sequential models benefit from the continuity and completeness of spectral features. These insights offer a strong validation for the proposed approach.

7) Effects of Equal or Unequal Subband Groups: In this study, we proposed degrouping bands into unequal subgroups and then applying CNNs to each group. While some previous works sliced spectral bands into equal subgroups, this could lead to unexpected feature mixing from uncorrelated adjacent bands. To verify the impact of this unequal degrouping strategy, we conducted experiments with specific hyper-parameter

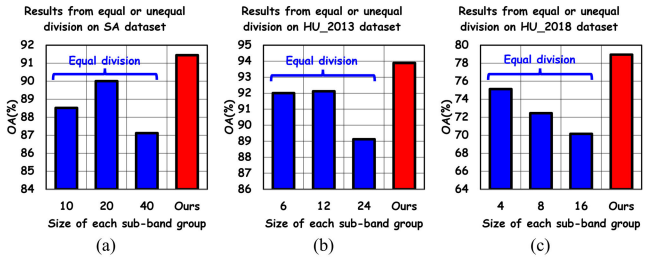


Fig. 19. Comparison between equal division (blue) on spectral bands and unequal division (red) on spectral bands. Ours are following the unequal division shown in Fig. 11. Experimental results on (a) SA dataset, (b) HU_2013 dataset, and (c) HU_2018 dataset.

settings, using channel lengths of 10, 20, and 40 for the SA dataset; 6, 12, and 24 for the HU_2013 dataset; and 4, 8, and 16 for the HU_2018 dataset. The results are presented in Fig. 19.

We found that our method of unequal spectral division achieved better results than equal division. Specifically, smaller equal divisions may result in a higher number of subimages or cubes, negatively affecting model performance due to increased computational complexity. Moreover, when the number of subband groups is approximately 10, performance tends to improve. For the datasets, this meant ten groups for SA, 12 for HU_2013, and 12 for HU_2018.

However, equal division alongside adjacent bands can cause unintended feature mixing among local bands, degrading the model's ability to process spectral features. This is likely why our method of unequal division consistently outperforms equal division. In addition, the fewer subband groups resulting from our method of unequal division make it more effective and lightweight for handling spectral features.

8) Exploring the Capacity With 3-D-Multiscannings: In light of the improved performance observed through the utilization of a 3-D-multiscanning strategy, it is natural to pose two important questions: 1) What is the capacity of the multiscanning process and 2) Which scanning is the most significant? To further explore these questions, we conduct experiments focusing on three key points as follows.

1) The impact of a single 3-D-scanning.
2) The saturation effect of utilizing multiple 3-D-scannings.
3) The relative importance of each 3-D-scanning.

In the first point, we experimented with using a single scan order for training. The results, as shown in Fig. 20(a)–(c), reveal that each individual scan produces a similar OA. This consistency may be due to the limited generative capacity inherent in using only one scan order. Moreover, the less-than-satisfactory validation accuracy observed in these experiments underscores the underfitting of the model when it is trained with a single scanning method.

In the second point, we carried out experiments using a varying number of scans, ranging from 1 to 8. It's noteworthy that for even numbers of scans (2, 4, 6, and 8), bi-directional processing was employed. The results are depicted in Fig. 20(d). We observed that increasing the number of scannings leads to improved classification accuracy, thereby affirming the efficacy of the multiscanning approach. However,

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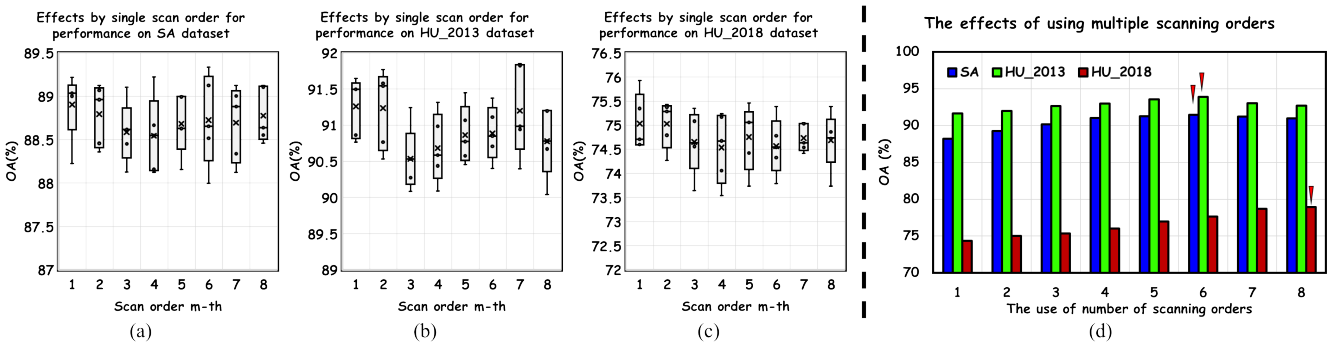


Fig. 20. Effects of using single and multiple scan orders for evaluating the performance. For each scan order, we have conducted experiments five times, as recorded, in (a) SA dataset, (b) HU_2013 dataset, and (c) HU_2018 dataset. The effects of using multiple scan orders for performance are shown in (d).

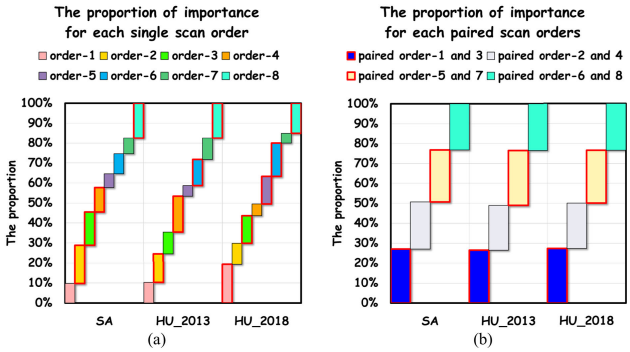


Fig. 21. (a) Importance proportion of each single scan order, as calculated by counting the frequency of each scanning-based weight being greater than 0.125 or not for different datasets. (b) Importance proportion of paired scan orders, as calculated by counting the frequency of each paired scanning-based weight being greater than 0.25 or not for different datasets.

a noticeable saturation in performance occurs beyond a certain number of scan orders, suggesting that including too many scans may lead to performance degradation. The optimal number of scans for the SA dataset was found to be 6. Similarly, for the HU_2013 and HU_2018 datasets, which are complex and characterized by unbalanced spatial distribution, performance saturation was noted at 6 and 8 scans, respectively.

In the third point, we evaluated the attention weights assigned to each 3-D scan order for feature fusion, as described in Section. II-F. We conducted numerous experiments to determine the significance of each 3-D scan order, as well as each pair of 3-D scan orders, for each dataset. This was done by counting the number of times their weights exceeded the average, as illustrated in Fig. 21. The experiments were split into two categories: 1) focusing on single 3-D scan orders and 2) focusing on paired 3-D scan orders. The results are presented in Fig. 21(a) and (b), respectively. In the single scanning analysis, we observed that the model adaptively assigns varying weights to each scan order for a given HSI patch. Some scan orders are assigned higher weights (i.e., greater than the average, highlighted by red squares), suggesting they positively influence model training, while others appear to have a less important impact. Moreover, the significance attributed to each scan order changes when multiple scans are combined and this variation is highly dataset-specific and dependent on the initial samples. However, across all datasets, scan order-8 consistently emerged as more important than others. In the paired scan analysis, we discovered that the combinations of scan orders 1 and 3, as well as 5 and 7, consistently received higher weights across all three datasets.

TABLE X
COMPARISON OF USING FEATURE-LEVEL FUSION AND OUTPUT-LEVEL FUSION FOR OA

	Feature-level fusion	Output-level fusion
SA	90.021	**91.455**
HU_2013	93.002	**93.884**
HU_2018	76.229	**78.947**

9) Exploring the Feature-Level and Output-Level Fusion: We developed our multiscanning feature fusion strategy in two distinct ways: 1) feature-level fusion and 2) output-level fusion. To evaluate the effectiveness of these two approaches, we conducted experiments across three datasets. The results of these experiments are detailed in Table X.

We observed that the output-level feature fusion approach tends to be more discriminative in scenarios where training samples are limited, imbalanced, and disjoint. In our experiments, the SA and HU_2018 datasets were tested using a small number of training samples, while the HU_2013 dataset employed disjoint samples provided by the GRSS. As a result, the samples across all three datasets exhibited significant disjointedness in their distribution. Under these conditions, output-level fusion of multiscanning features demonstrated superior performance compared to feature-level fusion. This finding offers valuable insights, particularly in contexts where multiple features must be analyzed with a limited sample size, such as in multiview learning with constrained data.

Furthermore, we posit that the enhanced generative capability of the output-level fusion strategy stems from each output feature representing a unique element within the feature space. The integration of multiple scan orders potentially enriches the feature pool, creating a buffer area in the feature space that facilitates a more robust feature representation.

10) Effects of the Number of Training Samples: Our objective is to explore how the number of training samples affects the performance of each method. The sole variable is the number of selected training samples, with all other parameters remaining constant. Given that the disjointed

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TABLE XI
TOTAL RUNNING TIME (MINUTES) AND THE NUMBER OF TRAINABLE PARAMETERS REQUIRED BY DIFFERENT METHODS ON THE SA DATASET

	General ViT [78]	SpeFormer [24]	HiT [27]	SSFTT [79]	3D-ViT [80]	CasRNN [21]	Multi-LSTM [50]	Multi-RT [51]	Ours
Train (mins.)	0.81	1.3	20.34	6.11	11.51	3.31	5.54	18.23	6.94
Test (mins.)	0.3	0.8	3.52	1.04	2.86	0.9	1.19	2.91	1.26
Parameters	141K	381K	45M	2.9M	5M	689K	1.1M	2.5M	1.4M

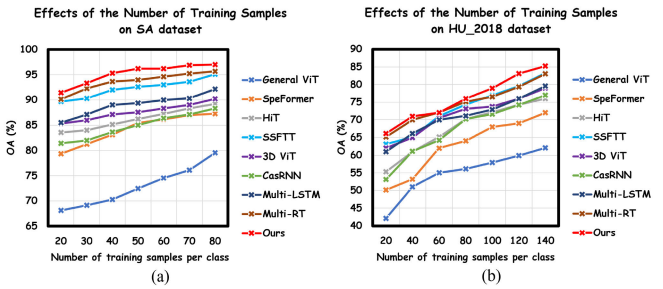


Fig. 22. OA of different models with different numbers of training samples. (a) SA dataset. (b) HU_2018 dataset.

samples in the HU_2013 dataset are predetermined by the GRSS community, we conducted experiments only on the SA and HU_2018 datasets, maintaining a fixed number of samples per class. For the SA dataset, the training sets consisted of 20, 30, 40, 50, 60, 70, and 80 labeled samples from each class. In contrast, the training sets for the HU_2018 dataset ranged from 20, 40, 60, 80, 100, 120, to 140 samples per class.

The results are depicted in Fig. 22. We observe that the OA of each method improves with an increase in the number of training samples. Our proposed method outperforms others even with fewer samples. This is particularly evident in the HU_2018 dataset, where the substantial improvements confirm that a sufficient number of training samples can enhance the model's generative capability. In the SA dataset, the performance improvement tends to be more gradual as the number of training samples increases. The rationale behind using a smaller number of samples for these experiments stems from real-world challenges in acquiring sufficient and balanced samples for such tasks, which can impede practical applications. Hence, achieving better results with fewer samples is a desirable outcome in this context.

11) Computational Costs: To compare computational costs among all methods, we have presented a statistical analysis in Table XI, adhering to the experimental settings described previously. This table records the total training time, testing time, and model size (number of parameters) for the SA dataset. All experiments were conducted on the Ubuntu 20.04 system with a GeForce RTX 3060 GPU.

Our method demonstrates competitive processing times, which are significantly lower than those of HiT, 3D-ViT, and Multi-RT. However, they are somewhat longer compared to General ViT, SpeFormer, Multi-LSTM, SSFTT, and CasRNN. In terms of trainable parameters, our method requires fewer parameters than those required by other state-of-the-art methods, indicating greater parameter efficiency.

Overall, our method provides a balanced tradeoff between running time and model complexity (as indicated by the number of parameters). While it does not boast the shortest training or testing times, it maintains a lower running time compared to several advanced methods, coupled with a relatively low parameter count. This balance suggests that our method could be an optimal choice for scenarios that demand efficiency alongside manageable model complexity.

IV. CONCLUSION

In this study, we introduce a novel approach that seamlessly integrates the strengths of convolution, recurrence, and self-attention for processing HSI patches in a 3-D configuration. Our method begins with a subband grouping-based group-wise convolution, designed for image dimension reduction while preserving spectral continuity for subsequent processing. This is followed by a cubic partition and an extended 3-D multiscanning strategy, which positions subcubes within the recurrence flow, considering multiple directions. The centerpiece of our approach is the SRT, which reimagines the ViT in a recurrent format, effectively combining spectral continuity and spatial contextual features in a 3-D framework. In addition, we propose a 3-D-multiscanning feature fusion technique for multidirectional feature aggregation. Our experiments on three public HSI datasets demonstrate that our method outperforms existing baselines and state-of-the-art methods, highlighting its effectiveness and efficiency in HSI classification. This research marks a significant step in the relatively unexplored territory of combining convolution, recurrence, and self-attention in HSI analysis. Looking forward, we aim to further refine our model by enhancing the integration of SRT with convolution units, hoping to contribute more to this field.

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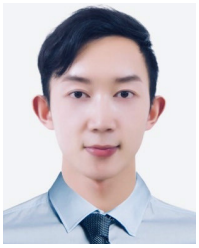
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